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Date: December 2023

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Numerical analysis of the dynamic behavior of arc by rotating laser-GMAW hybrid welding of T-joints

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Highlights

Outline

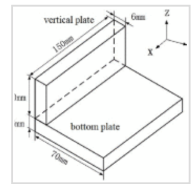
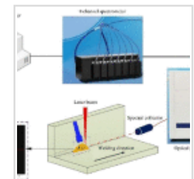
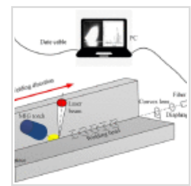
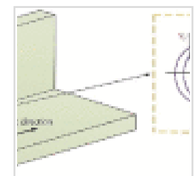
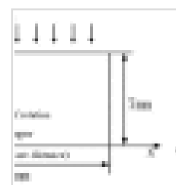
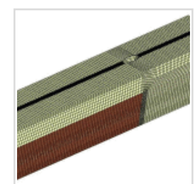
- A numerical analysis model is developed for the arc of hybrid welding with T-joint.
- The interaction behavior between arc and laser-induced metal vapor during welding process can be simulated.
- Laser-induced metal vapor has a cooling effect on the arc.
- The influence of laser rotation frequency on arc and laser-induced metal vapor is discussed.
- The influence of laser rotation radius on arc and laser-induced metal vapor is discussed.

Abstract

In order to improve the reliability and stability of complex joints hybrid welding, this paper establishes an arc numerical analysis model, studies the T-joint high-frequency rotating laser-GMAW hybrid welding arc physical characteristics, and expounds the effects of laser rotation frequency along with rotation radius on arc plasma temperature, velocity, and metal vapor concentration field distribution. The research results show: the arc is more likely to start on the bottom plate and vertical plate surfaces, and the current density at the angle between the plates decreases significantly; temperature drops at the hybrid welding arc tail, the current effective distribution radius was reduced, and current density was more concentrated; the laser beam rotation frequency increases, the eruption speed of the metal vapor ejected from the keyhole decreases, while the meeting height of arc plasma and laser-induced metal vapor decreases; and, with an increase in rotation radius, the laser rotation speed increases accordingly. At the same rotation frequency, the injected metal vapor kinetic energy and rising height are reduced.

Keywords

Figures (30)

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Tables (4)

Rotating laser; Hybrid welding; Dynamic arc behavior; Numerical simulation

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1. Introduction

Laser-arc hybrid welding organically combines the respective advantages of laser welding and arc welding, compensates for the shortcomings of both welding types, achieves fast welding speed, large penetration, and good stability and realizes high-efficiency welding. It is widely used in various industries with promising prospects in the area of industrial production. Complex joints such as T-joints fortify structures in nuclear power engineering, aerospace, military industry, ships, and other fields [1], [2], [3], [4], [5]. Due to the influence of weldment material physical properties, plate thickness, joint geometry, and welding position, the welding quality of complex joints is difficult to control [1], [6]. Compared with the traditional single heat source welding process, the high energy density laser is added and the corner joint obtains a larger penetration depth which solves the problem of non-penetration at the root of the complex joints welding seam. Compared with traditional laser welding, the oscillation of the high energy density laser can effectively reduce welding pores [7], [8], refine the weld grains, improve the welded joint toughness as well as the microstructure and properties of the joint itself [9], [10].

Fetzer et al. [11] observed the dynamic behaviors of the keyhole and bubble in the laser welding of aluminum alloys under different swing modes with X-ray imaging technology. The research shows that the laser rotation changes the coupling behavior of the bubble and keyhole, resulting in a decrease in molten pool bubbles. Moreover, it shows hoop rotation has a lower porosity than other swing modes. However, the study did not analyze internal mechanisms such as laser-induced plume, keyhole, bubble, and molten pool behavior. Based on the advantages of rotary laser welding, scholars began to add high-frequency rotary laser to the conventional laser-arc hybrid welding process and conducted preliminary research. Yang et al. [12] used the rotating laser for the narrow-gap rotating laser-MIG hybrid welding of 5A06 aluminum alloy. The results show that a suitable rotating laser can effectively reduce porosity. Due to the significant stirring effect of the rotating laser on the molten pool, the minimum grain size of joints is about 28% lower than that of joints welded without the rotating laser. Hao et al. [13] studied the oscillating laser-arc hybrid welding of AZ31B magnesium alloy. The results show that pore formation can be inhibited with an oscillating frequency higher than 75 Hz and a radius smaller than 0.5 mm. The neighboring columnar grains by the fusion line can be broken by the beam oscillation behavior while grain growth is promoted with an increase in frequency or radius. At present, researchers rely on high-speed camera information acquisition system, spectrometer information acquisition system as well as image processing technology and equipment to study arc plasma behavior parameters in the process of laser-arc hybrid welding as well as qualitatively analyze and infer relevant conclusions.

Liu et al.[14] studied the arc thermal distribution characteristics during ordinary pulsed MAG welding and Nd:YAG laser-pulsed MAG hybrid welding. The research results show that the electron temperature and electron density of ordinary pulsed MAG welding are higher than those of Nd:YAG laser-pulsed MAG hybrid welding; however, this study is based on the qualitative speculation of the detection results, and does not involve mechanism analysis such as laser-induced plume in the keyhole. Mahrle et al.[15] performed steel/aluminum laser-plasma arc hybrid welding experiments and numerical analysis simulations. It is found that when the laser power increases to a certain value, the metal vapor induced by the high-power laser will cause arc instability and a decrease in the arc axis current density. Gao et al. [16] tested and analyzed the rotating laser-GMAW hybrid welding stability. The study showed that the arc can be stabilized when the laser beam rotation parameters are appropriate. Chen et al. [17] used high-speed camera and image processing technology to study the carbon steel laser swing-MAG hybrid welding arc shape. They found that the rotating laser-arc pressure effect on the arc root was different from that of conventional hybrid welding.

Mu et al. [18] used high-speed camera equipment to demonstrate the arc plasma shape during hybrid welding under different laser-arc distance conditions. The research shows that the TIG welding arc shape is stable during the welding process while the laser-TIG hybrid welding arc shape fluctuates greatly at the tail and the shape of its bright white area is significantly smaller than that of TIG arc. In addition, as the distance between the laser beam and the end of the welding wire decreases, the arc topography average area gradually decreases while fluctuation increases. The study also found that the laser-TIG hybrid welding arc shape fluctuation area is mainly concentrated on the left side of the laser beam while the arc plasma profile on the right side of the laser beam is relatively stable, especially the area near the tungsten pole end.

In order to compensate for the lack of process test or detection, it is necessary to use numerical simulation technology to study physical mechanisms such as laser disturbance to the arc, laser-induced plume in the keyhole, and molten pool thermal distribution. Cho et al. [19] established a hybrid welding arc model and demonstrated the influence of laser power on the disturbance of arc plasma. It was found that the plasma current density and temperature within the vicinity of the laser irradiation area changed with the degree of laser power. Murphy et al. [20], [21] considered the arc deformation and workpiece surface, then established a two-dimensional GMAW welding numerical analysis model, demonstrating how the workpiece surface deformation affects the current density distribution which in turn affects the welding pool temperature field. Nevertheless, this model is only suitable for 2D surfacing process, and its arc characteristics are different from those of complex joint hybrid welding. Mahrle et al. [15] conducted experiments and simulation studies on laser-plasma arc welding. The theoretical results of numerical simulations show that arc plasma characteristics are not directly affected by laser radiation but by laser-induced metal evaporation. As the current density increases, arc stabilization is only suitable for medium-rate metal vaporization, and when the metal vapor rate exceeds a certain threshold, it causes arc instability such that the current density along the arc axis decreases. MA et al. [22] studied the hybrid laser-electric arc heat source welding of dissimilar Mg alloys through experiments and simulation. The research shows that a hybrid heat source can promote laser energy absorption and an electric arc in the molten pool, resulting in more uniformed energy distribution in the molten pool and a corresponding improvement in welding parameters. Pang et al. [23], [18] studied the interaction between laser and arc plasma in laser-arc hybrid welding and found that laser can assist in arc initiation and has a cooling effect on the arc plasma. Gao et al. [24] developed a 3D non-axisymmetric hybrid plasma model in fiber laser-arc welding, in which the laser induced plasma is considered as ejecting vapor from keyhole with specific radius, temperature and velocity. Through calculations, it was found that the temperature of the hybrid plasma decreases compared with the temperature of arc plasma under all the simulation parameters.

At present, researchers mainly focus on simple joint swing laser welding, while very few study T-joint rotary laser hybrid welding. Therefore, this paper intends to study the physical properties and transient behavior of the T-joint high-frequency rotating laser-GMAW hybrid welding plasma by means of numerical simulation and experimental detection, as well as to clarify the transient evolution process and coupling behavior characteristics of keyhole, thermal field and flow field, welding quality realize active control, and improve the complex joints hybrid welding reliability and stability. The main difference between this paper and previous research lies in the emphasis on the study of joint structure and the influence of rotating laser on arc plasma. It differs from research on simple joints in the past and is in contrast to research on non-rotating lasers.

2. Experimental procedure

The experimental plate is a 5083 aluminium alloy with a composition of 1.2 mm diameter welding wire as shown in the Table 1. The joint size and type are shown in Fig. 1. Welding was carried out using a fibre laser (YLS-6000W) and a GMAW welding machine (Fronius TPS4000). The laser-related parameters are shown in the Table 2. To study the influence of the laser rotation parameters on the arc behaviour, the rotation frequency and radius of the laser was varied. The welding was carried out with the laser in the lead, the angle between the wire and the laser was 30° and the initial distance between the wire end and the workpiece surface was 5 mm. the shielding gas was argon with a purity of 99%. The specific process parameters are shown in Table 3.

Table 1. Composition of welding materials (mass fraction /%).

Materials	Si	Fe	Mg	Cu	Mn	Cr	Zn	Ti	Al
5083	≤0.4	0.4	4.0~4.9	≤0.1	0.4~1.0	0.005~0.25	0.25	≤0.15	Bal.
ER4043	4.5~6.0	≤0.8	≤0.05	≤0.3	≤0.05	0.05~0.20	≤0.1	≤0.2	Bal.

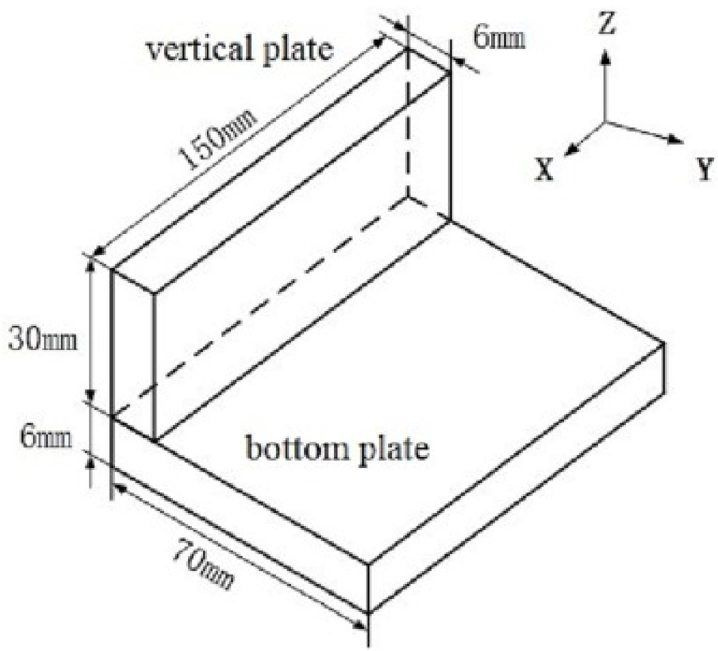

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Fig. 1. T-joint.

Table 2. Related parameters of the laser.

Laser head model	Fiber core diameter (μm)	Spot diameter (μm)	Focal length (mm)	Laser wavelength (nm)	Output wavelength (nm)	Operating mode
High YAG-BIMO	200	290	310	1075	1025～1080	Continuous output multimode

Table 3. Experimental parameters.

Process parameters	Value range
Laser power p (kW)	1/2/3
Welding speed v ($mm{\cdot}s^{-1}$)	15
Welding current I (A)	180
Arc and laser angle θ (°)	30
Defocus amount Δf (mm)	0
Distance between laser and arc d (mm)	2
Laser focus diameter Φ(mm)	0.2
Gas flow Q_{gas} (L/min)	20
Wire elongation(mm)	15

The experiments were carried out using a high-speed camera information acquisition system and a spectral information acquisition system to record the arc shape and spectral intensity. The spectral information collection wavelength range of the spectrometer analysis system is between 180 and 1100 nm. The spectrometer analysis system spectral information collection wavelength range is between 180 nm and 1100 nm. It consists of a spectrometer, an optical path amplification system, and an optical signal transmission fiber. The model is MX2500+as shown in Fig. 2. The figure also shows the location of the spectral signal collection points which are distributed in the direction of the wire axis and near-surface of the workpiece. There are seven points in total, scanned from 1 to 7, with a 1 mm lateral distance.

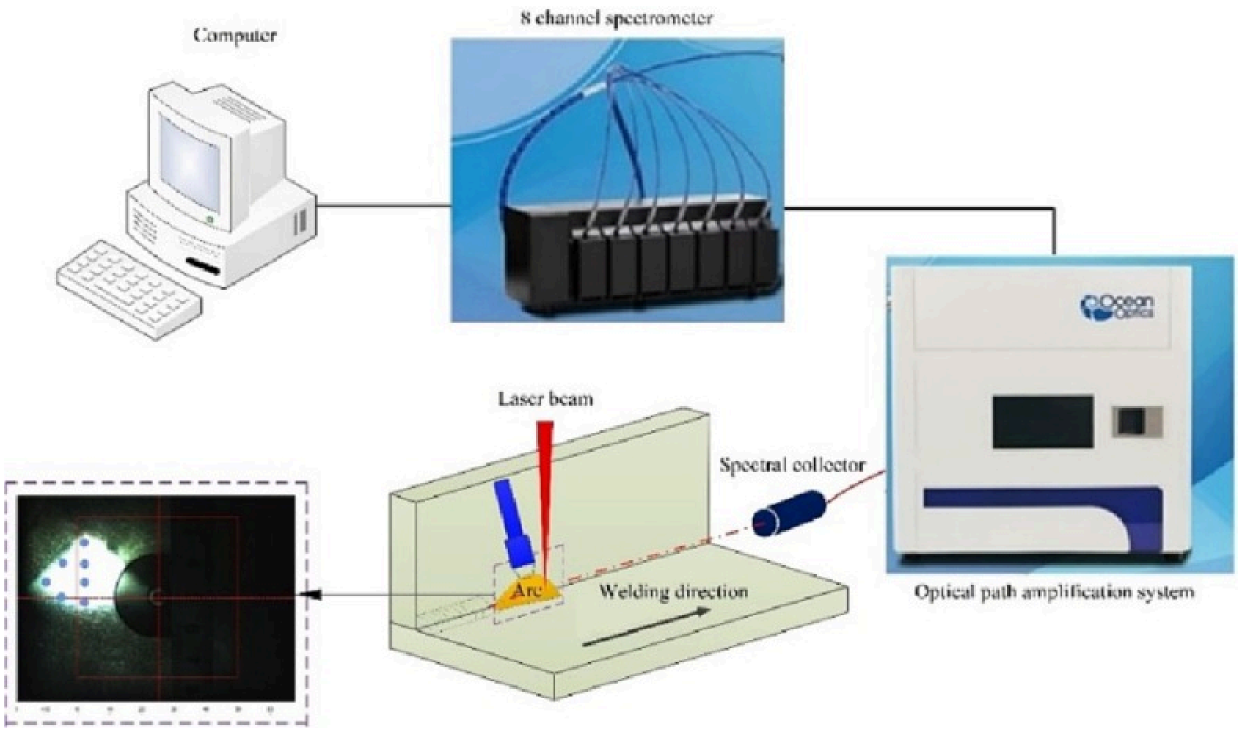

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Fig. 2. Laser plasma spectral analysis system and date-colleting platform.

The high-speed camera information acquisition system includes a high-speed camera system (FASTCAMSuper10K), DWETRON high-speed tester, auxiliary light source, and CP80-3-M540 camera. The lens has a built-in 808 nm filter that filters out the laser wavelength. During welding, the lens is pointed towards the weld seam, the torch and laser beam remain stationary, the self-made welding platform moves at a set speed, while the high-speed camera information acquisition system and the spectral information acquisition system record the arc shape and spectral intensity. After welding, the information captured by the high-speed camera system was used to determine the arc shape, while the spectral intensity and wavelength data were converted into the corresponding arc electron temperature for verification. Fig. 3 shows a schematic diagram of the device.

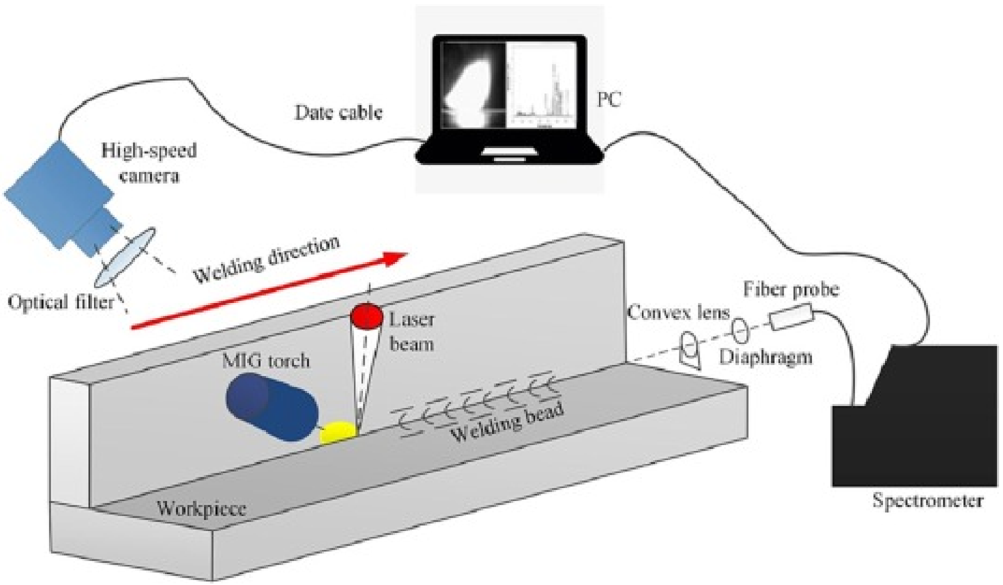

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Fig. 3. Schematic diagram of arc and spectrum information collection in T-joint.

3. Numerical modeling

3.1. Basic assumptions

The welding process involves complex MHD issues. Moreover, the physical processes involved in heat transfer as well as flow in the cathode and anode regions are also very complex. In order to simplify the model while highlighting the key points, this study will focus on laser-induced metal vapor effect on arc physical properties. The model will not consider the ionization energy and the dynamic behavior of the melt pool. Some reasonable assumptions and simplifications need to be made for the arc model. Assumptions of this model are as follows:

(1) The arc is an ideal Newtonian fluid, laminar and incompressible;

(2) The arc gas is pure Ar under atmospheric pressure, and the arc is in local thermodynamic equilibrium (LTE);

(3) The arc satisfies the optically thin property;

(4) Only the laser-induced metal vapor effect on the arc is considered;

(5) Customize functions of related thermodynamic and transport performance parameters with temperature as a variable;

(6) Deformation of the molten pool surface and droplets are not considered.

3.2. Governing equations[25]

Based on the aforementioned assumptions, basic governing equations such as mass, energy, momentum, electromagnetic field, and component transport are as follows:

Mass continuity equation is

$$\nabla \cdot (\rho v) + \frac{\partial \rho}{\partial t} = 0 \tag{1}$$

where ρ is the density and v is the velocity vector.

Momentum conservation equation is as follows:

The momentum equation in the x-axis direction is

$$\frac{\partial \rho v_x}{\partial t} + \frac{\partial}{\partial x}(\rho v v_x) = -\frac{\partial P}{\partial x} + \frac{\partial}{\partial x}\left[\left(2\mu\frac{\partial v_x}{\partial x}\right) - \frac{2}{3}\nabla \cdot v\right] + \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right)\right] + j_y B_z - j_z B_y \quad (2)$$

The momentum equation in the y-axis direction is

$$\frac{\partial \rho v_y}{\partial t} + \frac{\partial}{\partial y}(\rho v v_y) = -\frac{\partial P}{\partial y} + \frac{\partial}{\partial y}\left[\left(2\mu\frac{\partial v_y}{\partial y}\right) - \frac{2}{3}\nabla \cdot v\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y}\right)\right] + j_z B_x - j_x B_z \quad (3)$$

The momentum equation in the z-axis direction is

$$\frac{\partial \rho v_z}{\partial t} + \frac{\partial}{\partial y}(\rho v v_z) = -\frac{\partial P}{\partial z} + \frac{\partial}{\partial z}\left[\left(2\mu\frac{\partial v_y}{\partial y}\right) - \frac{2}{3}\nabla \cdot v\right] + \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z}\right)\right] + \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z}\right)\right] + j_x B_y - j_y B_x \quad (4)$$

where v_x, v_y, and v_z are the velocity vectors in the×,y, and z directions, respectively; P is the pressure; μ is the viscosity coefficient; j_x, j_y and j_z are the current density components induced by the magnetic field in the three directions, respectively; B_x, B_y, and B_z are the magnetic field strength components induced by the electric field in three directions, respectively. The electromagnetic force generated by the coupling of the electric field and magnetic field in the three directions drives the arc.

Energy conservation equation is

$$\frac{\partial \rho h_e}{\partial t} + \nabla \cdot(\rho v h_e) = \nabla \cdot\left(\frac{k}{c_p}\nabla h_e\right) + \frac{j_x^2 + j_y^2 + j_z^2}{\sigma_e} - 4\pi\zeta + \frac{5k_B}{2e} j \cdot \nabla T + q_{laser} \quad (5)$$

where c_p is the specific heat capacity; σ_e is the electrical conductivity; k_B is the Boltzmann constant; ζ is the net radiation coefficient. On the right side of the equation, the second term is the Joule heat generated by the current between the anode and cathode; the third term is heat lost to the outside world by radiation; the fourth term is the electron transfer heat generated by temperature difference; the fifth term is the energy absorbed by the laser-arc. Due to the short wavelength of the fiber laser, the arc plasma energy absorption will not be considered.

Electromagnetic field equation is given below:

Current continuity equation is

$$\frac{\partial}{\partial x}\left(\sigma_e\frac{\partial \phi}{\partial x}\right)+\frac{\partial}{\partial y}\left(\sigma_e\frac{\partial \phi}{\partial y}\right)+\frac{\partial}{\partial z}\left(\sigma_e\frac{\partial \phi}{\partial z}\right)=0 \tag{6}$$

Current density component is.

$$j_x=-\sigma_e\frac{\partial \phi}{\partial x},j_y=-\sigma_e\frac{\partial \phi}{\partial y},j_z=-\sigma_e\frac{\partial \phi}{\partial z} \tag{7}$$

where ϕ is the electric potential.

Magnetic field components are

$$\frac{\partial^2 A_x}{\partial x^2}+\frac{\partial^2 A_x}{\partial y^2}+\frac{\partial^2 A_x}{\partial z^2}=-\mu_0 j_x \tag{8}$$

$$\frac{\partial^2 A_y}{\partial y^2}+\frac{\partial^2 A_y}{\partial x^2}+\frac{\partial^2 A_y}{\partial z^2}=-\mu_0 j_y \tag{9}$$

$$\frac{\partial^2 A_z}{\partial z^2}+\frac{\partial^2 A_z}{\partial y^2}+\frac{\partial^2 A_z}{\partial x^2}=-\mu_0 j_z \tag{10}$$

$$B_x=\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}B_y=\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}B_z=\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y} \tag{11}$$

where A_x, A_y and A_z are the magnetic vector components in the three directions of×,y and z respectively; μ_0 is the vacuum permeability.

Component transport equation is

$$\frac{\partial}{\partial t}(\rho f_v)+\nabla\cdot(\rho v f_v)=\nabla\cdot(\rho D\nabla f_v) \tag{12}$$

where f_v is the mass fraction of metal vapor; D is the diffusion coefficient which is determined by the viscosity approximation equation is

$$D = \frac{2\sqrt{2}(1/m_1+1/m_2)^{0.5}}{\left\{\left(\rho_1^2/\beta_1^2\eta_1^2 m_1\right)^{0.25}+\left(\rho_2^2/\beta_2^2\eta_2^2 m_2\right)^{0.25}\right\}^2} \tag{13}$$

where m_1 and m_2 are the molar masses of aluminum and Ar, respectively; ρ_1 and ρ_2 are the densities while η_1 and η_2 are the viscosities of aluminum metal vapor and Ar, respectively; β_1 and β_2 are dimensionless constants.

In order to obtain a more accurate laser-induced metal vapor model during the simulation process. The morphology of keyhole under different laser power was captured by high speed camera experiment. According to the keyhole radius and laser energy density, the eruption temperature under different laser power was estimated by using the fitting relation of the laser-induced plasma model of Dilthey U[26]. The fitting relation was expressed as:

$$T = 0.00425 \times I + 3240.38462, 0.20mm < R \leq 0.30mm \tag{14}$$

$$T = 0.00431 \times I + 3252.13675, 0.30mm < R \leq 0.40mm \tag{15}$$

$$T = 0.00438 \times I + 3263.88889, 0.40mm < R \leq 0.50mm \tag{16}$$

where T is the plasma temperature; I is the laser energy density; R is the radius of the keyhole.

According to the metal vapor flow model in the area around the keyhole established by Amara [27], the ejection velocity at different laser powers is calculated as follows:

$$V_g = \sqrt{\kappa \frac{LT_k}{m_a}} \tag{17}$$

Where κ is the specific heat ratio; L is the ideal gas constant; T_k is the surface temperature of the keyhole; m_a is the atomic mass.

Rotation path is calculated by

$$\begin{cases} x(t) = x_0 + a \cdot \cos(2\pi \cdot f \cdot t) \\ y(t) = a \cdot \sin(2\pi \cdot f \cdot t) \end{cases} \tag{18}$$

where a is the laser rotation radius, f is the frequency, and x_0 is the laser rotation center position on the x-axis. The actual moving path of the laser heat source center shown in Fig. 4 needs to be described appropriately and accurately. Therefore, its rotation path formula refers to relevant literature [28] as shown above.

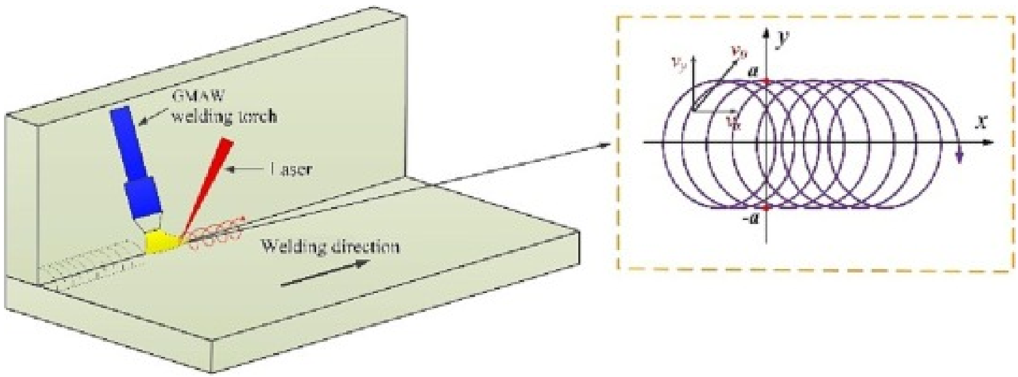

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Fig. 4. Laser rotation trajectory diagram.

3.3. Boundary conditions

The detailed boundary condition settings of the model are shown in Table 4.. In the table, r refers to the distance from the center of the laser beam spot while R is the keyhole eruption radius. The laser-induced metal vapor region boundary is determined by the following: if it is inside the keyhole region (r≤R), the plasma eruption velocity is V_g, the plasma temperature plasma is T, and the mass fraction of laser-induced metal vapor is 100%; if it is outside the keyhole area (r>R), the aluminum vapor eruption velocity is zero and the temperature is 1000K. Meanwhile, Ar ionization requires an initial temperature above 8000K. The surface potential of the vertical plate and bottom plate workpiece is zero, while the magnetic vector gradient is zero.

Table 4. Boundary conditions of the model.

Boundary	***F***	***V*(m·s^{-1})**	***T*(K)**	***φ*(V)**	***A*(Wb·m^{-1})**
Inlet	0	V_a	1000	$\frac{\partial \varphi}{\partial n}=0$	$\frac{\partial A}{\partial n}=0$
Outlet	0	$\frac{\partial V}{\partial n}=0$	1000	$\frac{\partial \varphi}{\partial n}=0$	0
Wiretip	0	0	3000	J_Z	$\frac{\partial A}{\partial n}=0$
Wirewall	0	0	1000	$\frac{\partial \varphi}{\partial n}=0$	$\frac{\partial A}{\partial n}=0$
Workpiece	1, $r \le R$ 0, $r>R$	V_g, $r \le R$ 0, $r>R$	T_g, $r \le R$ 1000, $r>R$	0	$\frac{\partial A}{\partial n}=0$

The current density set at the end of the wire is given as follows:

$$J_Z = \frac{I}{\pi r_w^2} \tag{19}$$

where I is the welding current and r_w is the wire radius.

3.4. Solution method

The calculation area longitudinal length is 25mm, the initial arc length is 5mm, the air layer height is 2mm, and the inclination angle of the included angle between the welding wire and weldment is 60°. The upper surface of the model is set as the shielding gas inlet, the workpiece surface (that is, the lower surface of the model) is set as the anode (Anode), and the remaining surfaces are set as the pressure outlet. The welding wire end is set as the cathode (Cathode) while its remaining surface is set as the wall (Wall). The x-axis positive direction is the welding direction and the z-axis positive direction is perpendicular to the workpiece, as shown in Fig. 5.

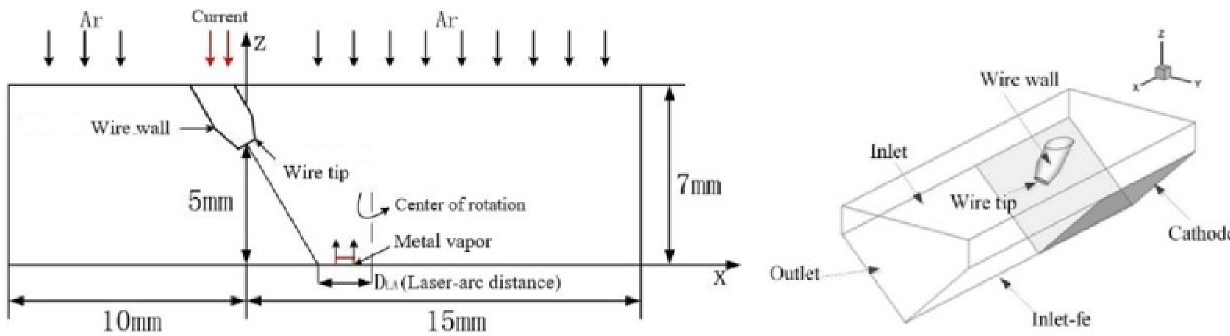

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Fig. 5. The sketch of the computational domain.

Based on the reference literature [29], [30], [31], it is known that the angle between the laser and arc can affect the interaction between the two. When the angle is too large, the inducing effect of the laser on the arc is weakened. Thus, it was decided to use a laser-arc angle of 30° for optimal results.

To simplify the calculation, the model is non-uniformly meshed, as shown in Fig. 6. The mesh near the welding wire should be smaller than 0.2 mm. The laser-induced metal vapor eruption region has a more severe disturbance to the arc plasma wherein a denser grid is used with a minimum size of 0.1 mm. Meanwhile, the remaining calculation region uses a sparser grid with a maximum size of 0.4mm, and a total of 273,854 nodes. In addition, the non-uniform time step is used in the calculation and the minimum time step is 10^{-6} s.

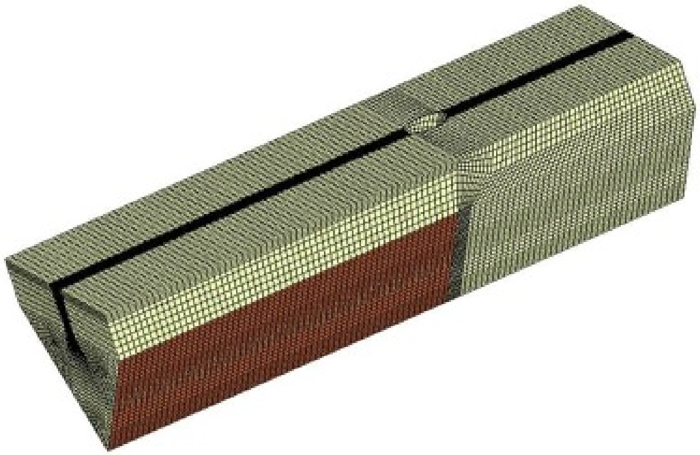

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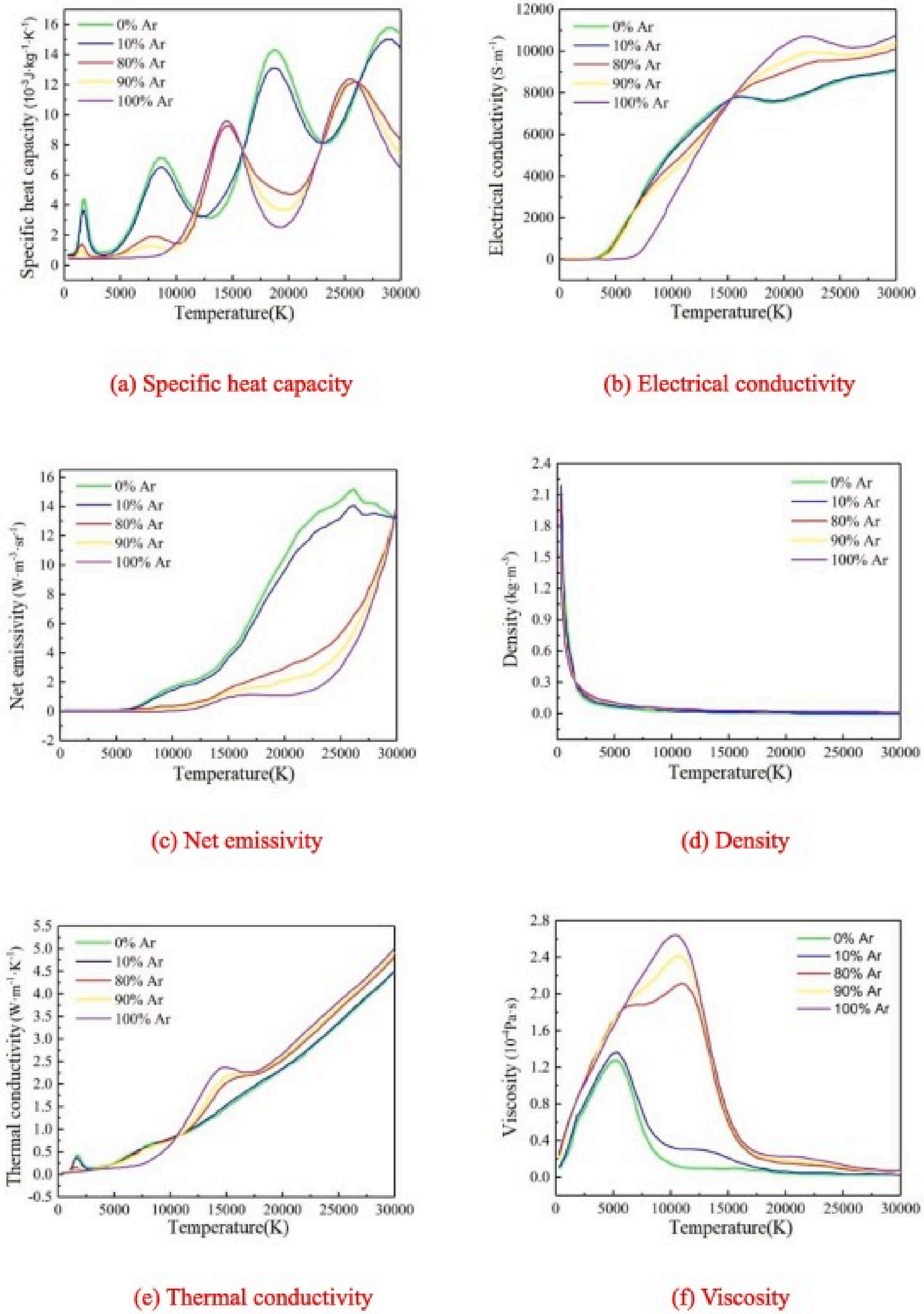
Fig. 6. The schematic of grid for arc model.

3.5. High temperature properties of materials

When studying the plasma characteristics in the welding process of composite heat source, only the effect of metal vapor induced by laser on arc plasma is considered. Arc region is Ar as a protective gas. During the process of Ar ionization and discharge at high temperature to arc formation, its various physical property parameters will change, such as density, thermal conductivity, conductivity, viscosity and radiation coefficient, etc. These parameters will also affect the dynamic characteristics and interaction effect of composite plasma. Fig. 7 shows the thermal physical performance parameters [32] of 100%Ar, 90%Ar and 10%Al, 80Ar and 20%Al, 10%Ar and 90Al and 100%Al at different temperatures, and the parameter values of other ratios are solved by linear interpolation method[32], [33], [34], [35], [36].



(a) Specific heat capacity

(b) Electrical conductivity

(c) Net emissivity

(d) Density

(e) Thermal conductivity

(f) Viscosity

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Fig. 7. Mixed arc plasma physical parameters [32].

4. Results and discussion

During the T-joint hybrid welding process, due to the rotation of the laser beam, the direction and size of the laser-induced metal vapor flow may change, causing the arc to be affected by different degrees of airflow interference at different positions. The temperature, pressure, density and other parameters of the arc gas may undergo asymmetric changes on different sides due to the rotation of the laser beam, thereby resulting in the asymmetry of the physical field of the arc.

Due to the arc shape asymmetry, the longitudinal section was taken at S1, S4, and S5 in the analysis, as shown in Fig. 8. The three sections are parallel to each other, and the included angles with the vertical plate and bottom plate are all 45°. For the cross-section, the S2 section at the welding wire axis and the S3 section at the keyhole are selected. Then, the arc characteristics on the longitudinal section (Y=0) when the laser beam is located at the four typical positions of A, B, C, and D are selected-. The rotation direction is A-B-C-D-A. The B and D positions are located at the weld central axis which are the two positions farthest and closest to the welding wire, respectively. The A and C positions are located on both sides of the weld, and the distance is the rotating laser radius.

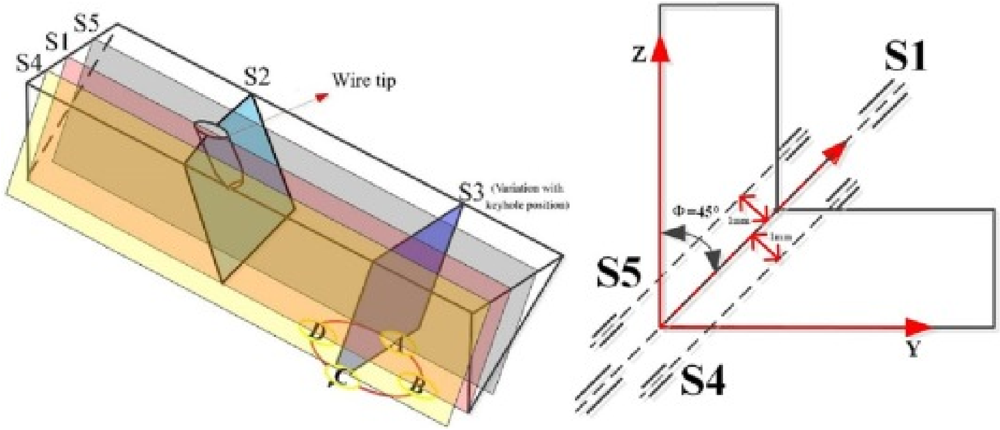

Fig. 8. The laser position at different times.

The study investigates the effect of laser rotation on arc behavior. While keeping all other parameters constant, the laser rotation frequency and radius are varied separately. The rotation frequencies are 50 Hz, 100 Hz, and 150 Hz, respectively, and the rotation radii are 0.5 mm, 1 mm, and 2 mm, respectively.

4.1. Comparison of temperature distribution between hybrid welding and GMAW welding

Fig. 9 shows the temperature distribution of the hybrid welding plasma and the GMAW welding plasma on the longitudinal section (S1). It is observed that, for both the processes, the peak temperature of arc plasma is generated at the region near the electrode tip. With distance from electrode tip along the central line, the temperature decreases shapely. The basic distribution feature is consistent with those in previous work[37], [38], indirectly demonstrating the reasonability of the established model.

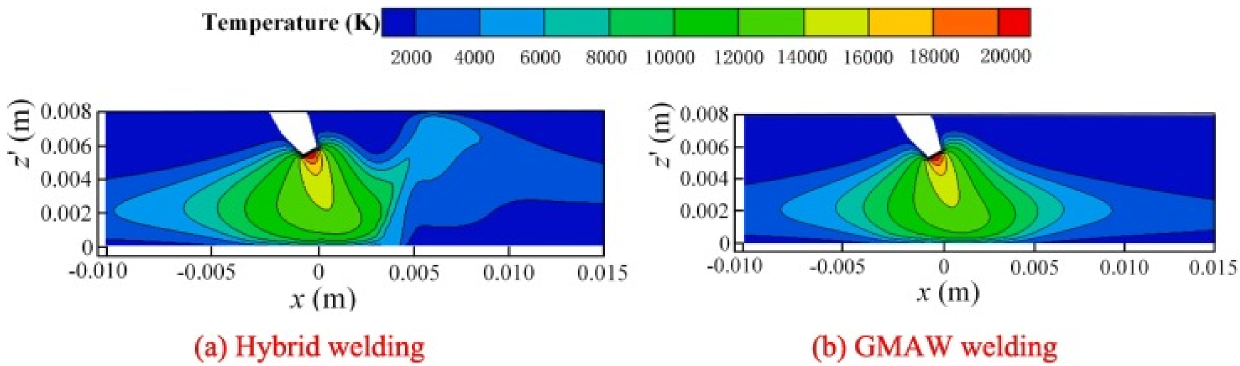

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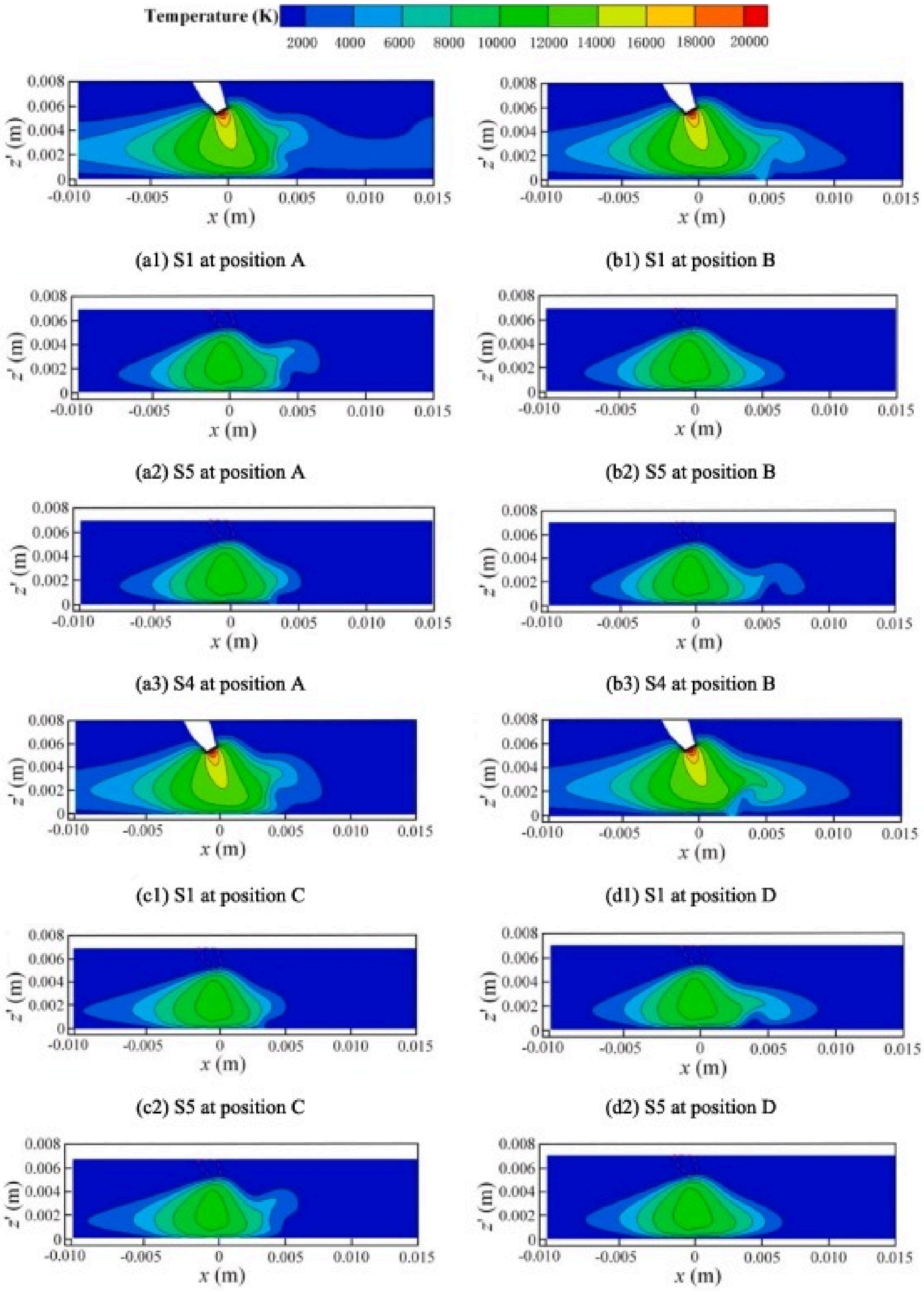
Fig. 9. Temperature field distribution of plasma on longitudinal section (S1).

In Fig. 9(a), it is seen that after the laser is added, the metal vapor ejected from the keyhole at a high speed in the laser induced metal vapor action area, causing arc plasma disturbance, and the temperature at the arc tail (metal vapor action area) changes significantly. The temperature distribution characteristics of the two welding methods are similar on the left side of the laser beam, and the highest temperature of both methods appears near the end of the welding wire. When no laser is added, the arc is in a state of free divergence. After the laser is added, the arc in the metal vapor acting area changes from a state of free divergence to a position confined to the keyhole, and the arc energy distribution is more concentrated. Compared with the same position temperature(12064K) at the tail of GMAW welding arc (metal vapor affected region), the plasma temperature(7670K) above and to the right of the laser radiation is significantly reduced due to the disturbance of laser-induced metal vapor. Therefore, laser has the effect of cooling arc.

4.2. Typical physical quantity distribution characteristics

4.2.1. Temperature distribution

Fig. 10 shows the effect of rotating the laser on the arc plasma temperature distribution when the laser beam is at different positions. The laser beam keeps moving according to the rotation path, and the high-speed laser-induced metal vapor and in the keyhole constantly changes the disturbance position of the arc plasma so the hybrid welding arc shape is always changing. The main changes in the temperature distribution characteristics appear at the tail of the arc. Also, the position where the arc root is constrained varies with the keyhole position.



(a1) S1 at position A

(b1) S1 at position B

(a2) S5 at position A

(b2) S5 at position B

(a3) S4 at position A

(b3) S4 at position B

(c1) S1 at position C

(d1) S1 at position D

(c2) S5 at position C

(d2) S5 at position D

(c3) S4 at position C

(d3) S4 at position D

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Fig. 10. Longitudinal section temperature field at different positions(H=100Hz, r=1mm).

At section S1, the arc topography is almost the same when the laser beam is rotated to the A and C positions. This is because the distance between A and C relative to the end of the welding wire is the same, and the difference in the degree of interaction between the laser-induced metal vapor and arc plasma is small, as shown in Fig. 10(a) and Fig. 10(c). Affected by the geometric characteristics of the joint, when the laser beam rotates to position C on the bottom plate, the laser-induced metal vapor kinetic energy is partially offset by gravity, and the meeting height is 1.7mm. Conversely, when the laser beam rotates to position A on the vertical plate, the laser-induced metal vapor kinetic energy will be supported by gravity, and the meeting height will increase to 1.9mm. In addition, the laser compresses the arc more at the C position than when it is at position A. When the laser beam is at point B, the laser-arc distance is largest while the meeting height of the arc plasma and laser-induced vapor plume is highest at 3.15mm as shown in Fig. 10(b). However, when the laser beam is at point D, the laser-arc distance is smallest while the meeting height is 1.5mm as shown in Fig. 10(d).

The metal vapor flows upwards with high velocity from the keyhole opening. Due to the effect of arc plasma impacting downwards, the metal vaper is forced to deflect at a certain height. The location where the vapor just starts to defect, is regarded as the meeting height.

When the laser beam is rotated to the A or C position, there is an arc topography asymmetry for the symmetrical sections S4 and S5. When the laser beam rotates to the D position, there is a clockwise upward rotating tangential force and the laser-induced metal vapor will deflect to the S5 section so that the S5 section arc tail is disturbed by the laser-induced metal vapor. When the laser beam rotates to the B position, there is a clockwise downward rotating tangential force and the laser-induced metal vapor will deflect to the S4 section so that the S4 section arc tail is disturbed by the laser-induced metal vapor. During the process of rotating laser hybrid welding, the metal surface is irradiated by the laser beam, which melts and evaporates, forming a metal vapor cloud surrounding the arc. As the rotating laser causes the metal vapor cloud to rotate along the welding axis, the motion state and speed of the metal vapor are constantly changing, forming a movement similar to an eddy current. This movement pushes the metal vapor cloud in the direction of rotation. Since the arc is located within the metal vapor cloud, it is also affected by this thrust, creating a tangential force on the arc.

Fig. 11 shows the S3 cross-section arc temperature distribution when the laser beam is in different positions. It can be seen from the figure that the laser-induced metal vapor has a cooling effect on the arc. When the laser beam is in different positions, the cooling area of the arc and the cooling effect are different. When the laser beam is at positions A and C, the areas of the laser-induced metal vapor cooling arc are above the vertical plate and above the bottom plate, respectively, as shown in Fig. 11(a) and Fig. 11(c). Compared with the D position, the B position arc cross-section is farther from the arc center and the arc temperature is lower so that the laser-induced metal vapor has a better cooling effect on the arc as shown in Fig. 11(b) and Fig. 11(d). This may be related to the different dynamic behaviors of laser-induced metal vapors at different locations.

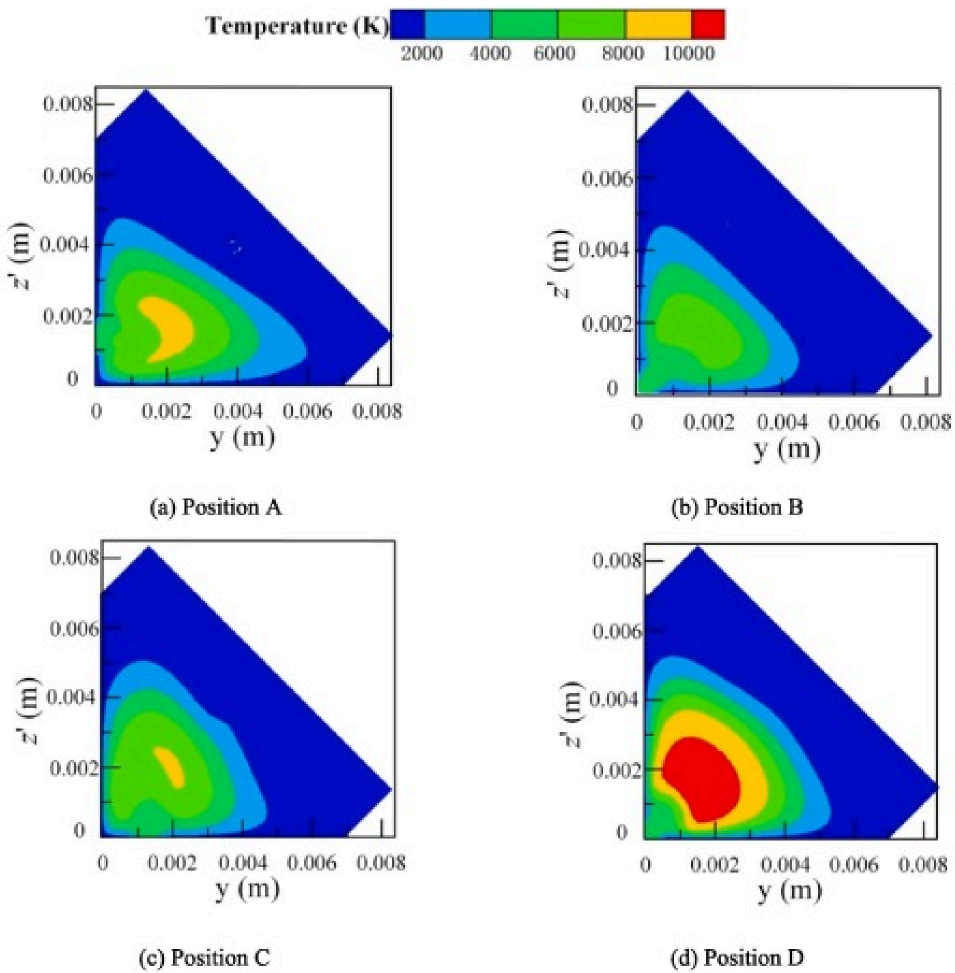

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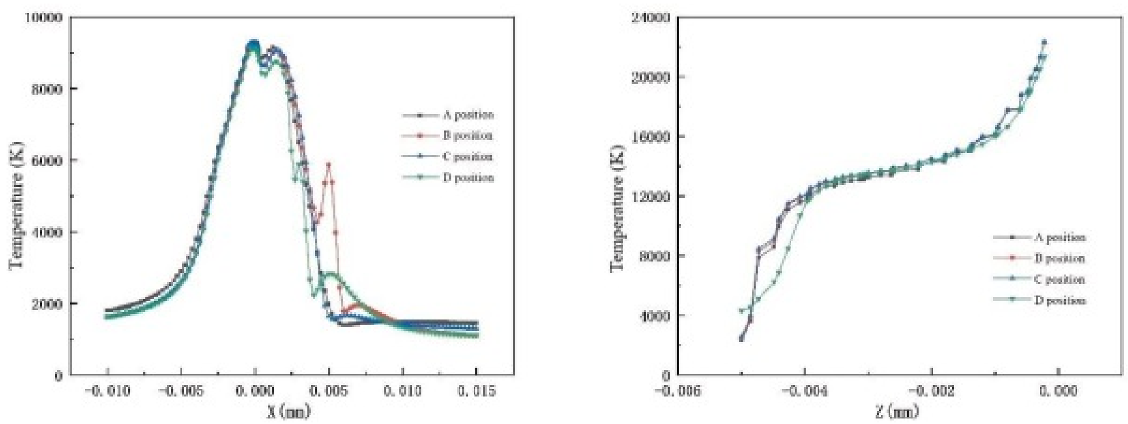
Fig. 11. Cross-section (S3 section) temperature field at different positions(H=100Hz, r=1mm).

Fig. 12 shows the arc temperature distribution on the workpiece surface at the S1 section and along the wire axis when the laser beam is at different positions. When the laser beam rotates to the four positions of A, B, C, and D, the highest temperature appears near the end of the welding wire, while the peaks are similar at 22178K, 22176K, 22182K, and

22169K, respectively. Comparing the workpiece near-surface temperature when the laser beam is at the four positions, it is found that the arc temperature distribution in the left area of the arc center is roughly the same, while the metal vapor region temperature is much higher when the laser beam on the right side rotates to the B position. This is because the laser action area energy is higher, the metal vapor has a higher conductivity, the arc root is compressed, and the hybrid welding arc plasma is more concentrated. When the laser beam rotates to the D position, close to the end of the welding wire, the laser-induced metal vapor disturbs and cools the arc tail more apparently. Therefore, when the laser beam is at the D position, the workpiece near-surface temperature by the keyhole in the S1 section is small.



(a) Surface of the workpiece

(b) Along the axis of the welding wire

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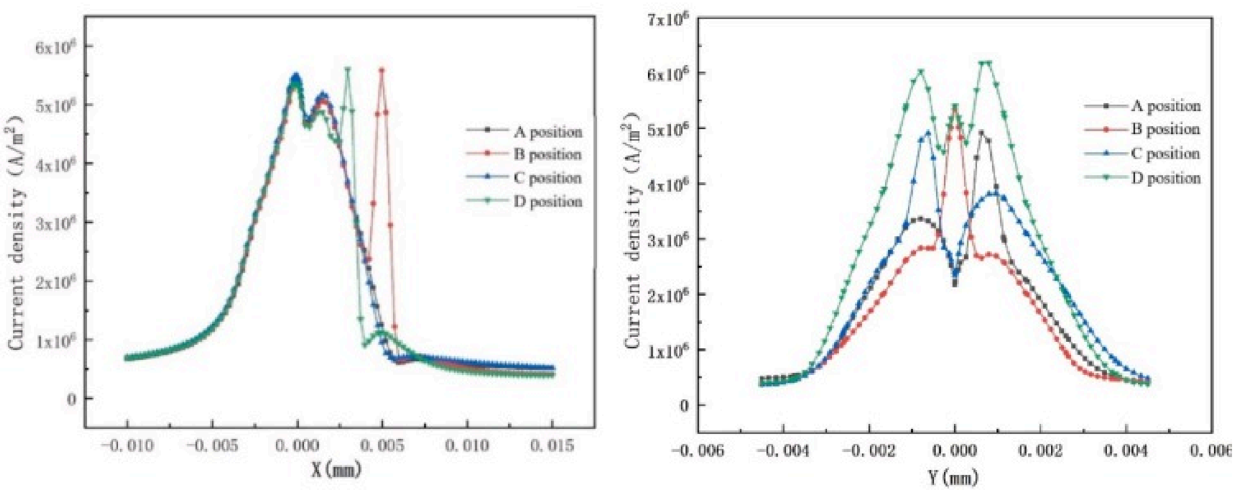
Fig. 12. Workpiece surface temperature distribution at the S1 section at different positions(H=100Hz, r=1mm).

4.2.2. Current density

Fig. 13 shows the current density curves along the x-axis and y-axis of the workpiece surface at different positions when the rotation frequency is 100Hz and the rotation radius is 1 mm. The compressed position of the arc root will move with the rotation path, and the effective distribution radius of the arc will also change from time to time. When the laser beam is at position B, it is far from the welding wire end, the cross-section current density is smallest, and the laser-induced metal vapor causes the current density in this area to abruptly change. When the laser beam is at the D position, it is closest to the welding wire end, and the cross-sectional area current density is larger than that of the B position. Therefore, the farther the laser beam is from the arc center, the lower the current density. After interaction between the arc plasma and laser-induced metal vapor, the phenomenon of sudden increase in the current density above the keyhole is weakened. Observing the current density plots along the y-axis near the workpiece surface, it is found that, unlike a simple joint, the current density near the angle between the base plate and vertical plate drops abruptly regardless where the laser beam is rotated. Due to the influence of these characteristics, during the T-joint welding process, the welding wire end is closer to the workpiece surface of the bottom plate and vertical plate, the arc deviates from the welding wire axis, the space arc at the junction of the vertical plate and bottom plate does not function, while the current density is reduced.



(a) Along the x-axis

(b) Along the y-axis

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Fig. 13. Current density distribution of the workpiece surface at different positions(H=100 Hz, r=1 mm).

Combining the current density with the temperature distribution, the effect of metal vapor on arc disturbance is weakened when the laser rotates far away from the arc, and the current density is relatively small. The compression effect of laser on arc decreases and arc radius increases slightly. When the laser rotates close to the arc, the metal vapor has a greater effect on the arc, and the current density generally increases. The compression effect of laser on arc is greater, and the arc radius is slightly reduced. This can be demonstrated by comparing the temperature distribution at positions B and D. However, position A and C are the same distance from the arc, and the arc radius has little difference.

4.2.3. Concentration field distribution of metal vapor

Fig. 14 shows the concentration field distribution of metal vapors at different laser beam positions. When the laser beam is in the A or C position, the metal vapor is ejected from the bottom plate or vertical plate workpiece surface. Meanwhile, when the laser beam is in the B or D position, the metal vapor is ejected vertically upward from the bottom of the joint. Compared with the D position, the B position is further from the welding wire axis. The farther the metal vapor eruption position is from the arc center, the greater the eruption height because of a lower arc plasma velocity with high-speed moving arc plasma as well as less inhibition of metal vapors. Combined with the fact that the arc tail is affected by laser-induced metal vapor, it is again demonstrated that the cooling of the arc tail is related to the laser-induced metal vapor dynamic behavior.

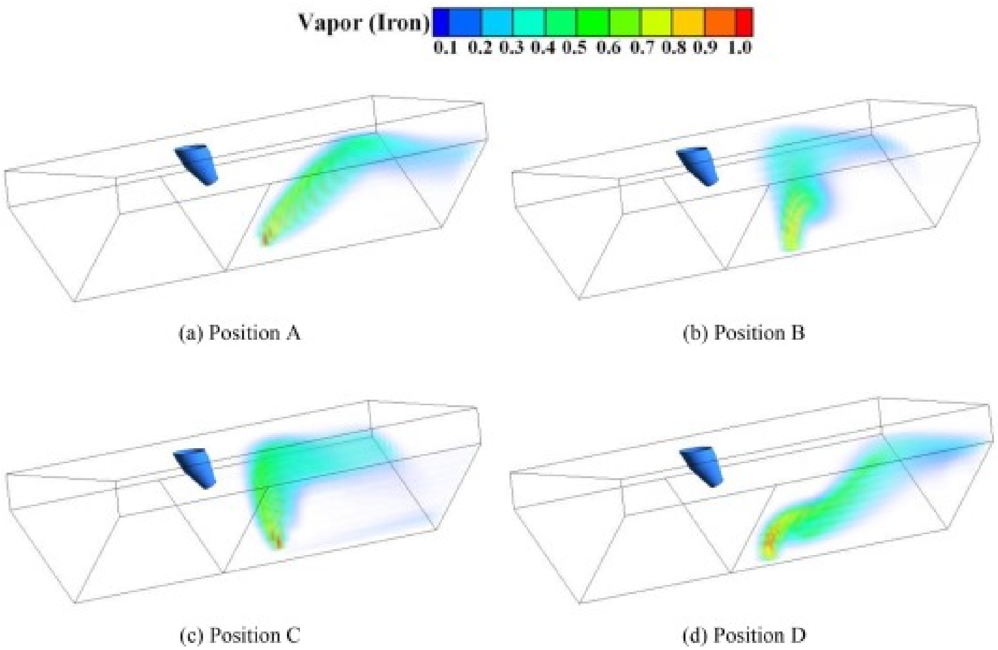

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Fig. 14. Concentration field distribution of metal vapor at different positions(H=100Hz, r=1mm).

Fig. 15 shows the metal vapor cloud diagram of a rotating laser with a rotating frequency of 100Hz and a rotating radius of 1mm at different positions. At position B or D, the keyhole is located at the S1 section, and the angle between the bottom plate and vertical plate in the laser-induced metal vapor spray area. Therefore, the laser-induced metal vapor disturbance area to the arc tail is in the area just above the angle between the bottom plate and vertical plate as shown in Fig. 15(b) and Fig. 15(d). At the A or C position, since the keyhole is on both sides of the S1 section, the arc tail oscillation comes from the influence of either the metal evaporated from the bottom plate or the vertical plate, as shown in Fig. 15(a) and Fig. 15(c).

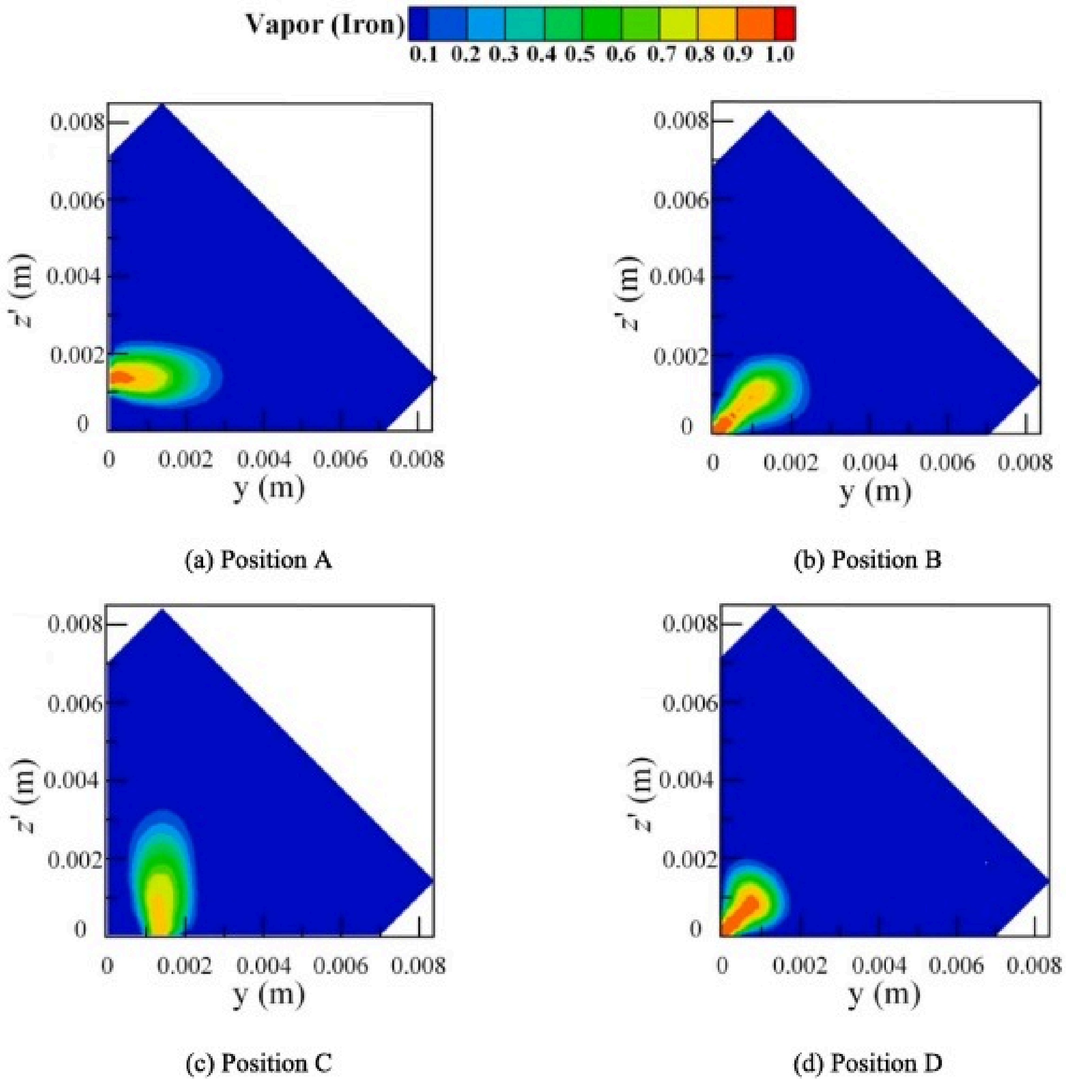

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Fig. 15. Metal vapor distribution at different positions(H=100 Hz, r=1 mm).

4.2.4. Velocity field

Fig. 16 shows the arc transient velocity field cloud diagram when the rotating laser with a rotating frequency of 100 Hz and a rotating radius of 1 mm is at different positions. When the laser beam is at the four positions of A, B, C, and D, the arc plasma flow velocity peak value all appear at the welding wire end which are 221.64 m/s, 218.97 m/s, 221.61 m/s, and 219.85 m/s, respectively. The peak difference is small because the laser-arc distance is 2 mm, the rotation radius is 1 mm, and the keyhole radius is 0.3 mm. The position of the laser-induced metal vapor injection is still a certain distance from the end of the welding wire which has little effect on the flow field near the welding wire end. After the high-frequency rotation of the laser, the laser-induced metal vapor jet kinetic energy decreases greatly while the encounter height also decreases.

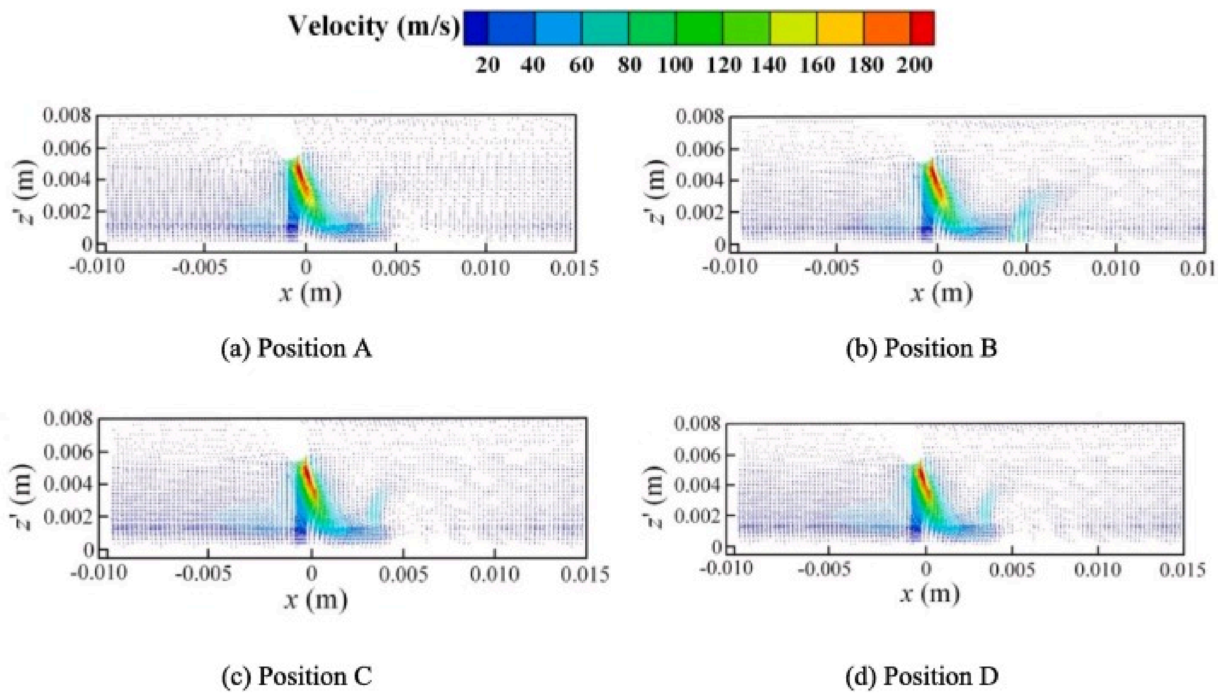

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Fig. 16. Velocity field at different positions(H=100 Hz, r=1 mm).

4.3. Effect of rotation frequency

4.3.1. Temperature distribution

As shown in Fig. 17 and Fig. 18, when the laser beam is at position B and the rotation frequencies are 50 Hz, 100 Hz, and 150 Hz, the arc temperature field maximum values are 22164 K, 22176 K, and 22181 K, respectively. The higher the rotation frequency, the higher the arc temperature peak. This is because the increase in rotation frequency of the laser beam increases its moving speed which in turn increases the chance of the laser beam approaching the arc plasma at the end of the welding wire per unit time as well as help improve the arc plasma and laser-induced plume interaction. It can also provide more charged particles for the arc to promote its ionization, thereby increasing arc temperature. When the rotation frequency is 50 Hz, 100 Hz, and 150 Hz, the maximum temperature of the cross-section where the B position is located is 7256 K, 7672 K, and 8045 K, respectively, while the arc peak temperature at the cross-section rises. This is because the laser beam cooling effect on the arc per unit time is weakened due to the increase in rotation frequency.

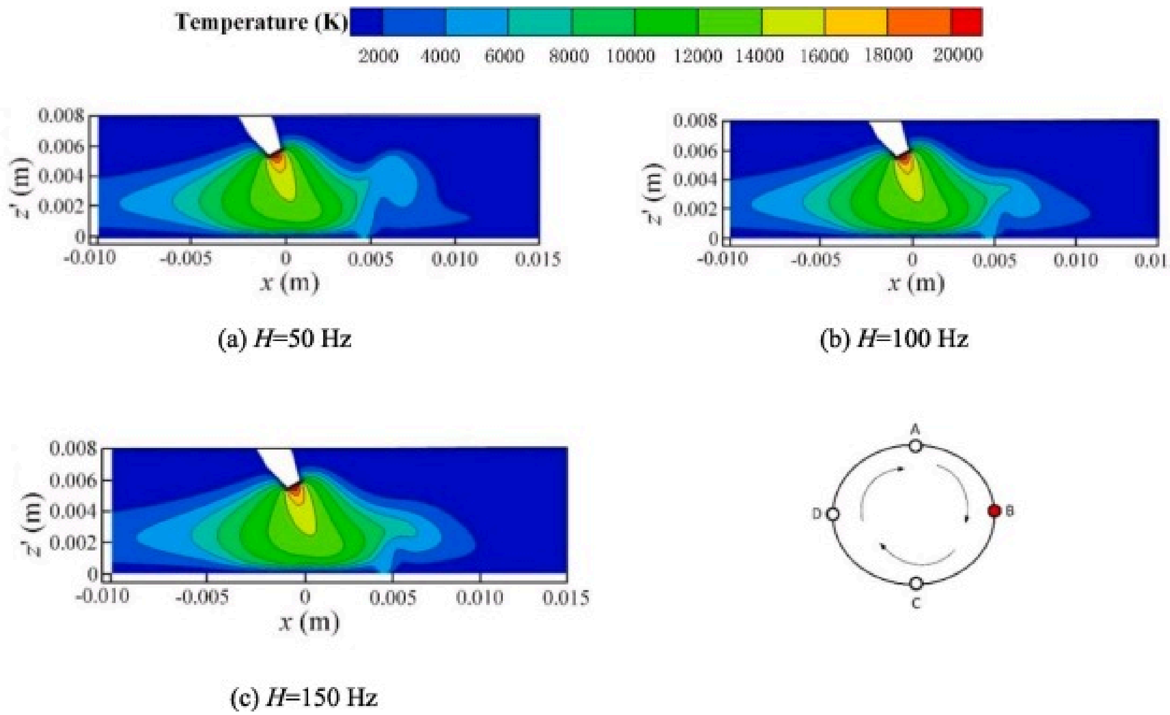

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Fig. 17. Longitudinal section temperature field at position B.

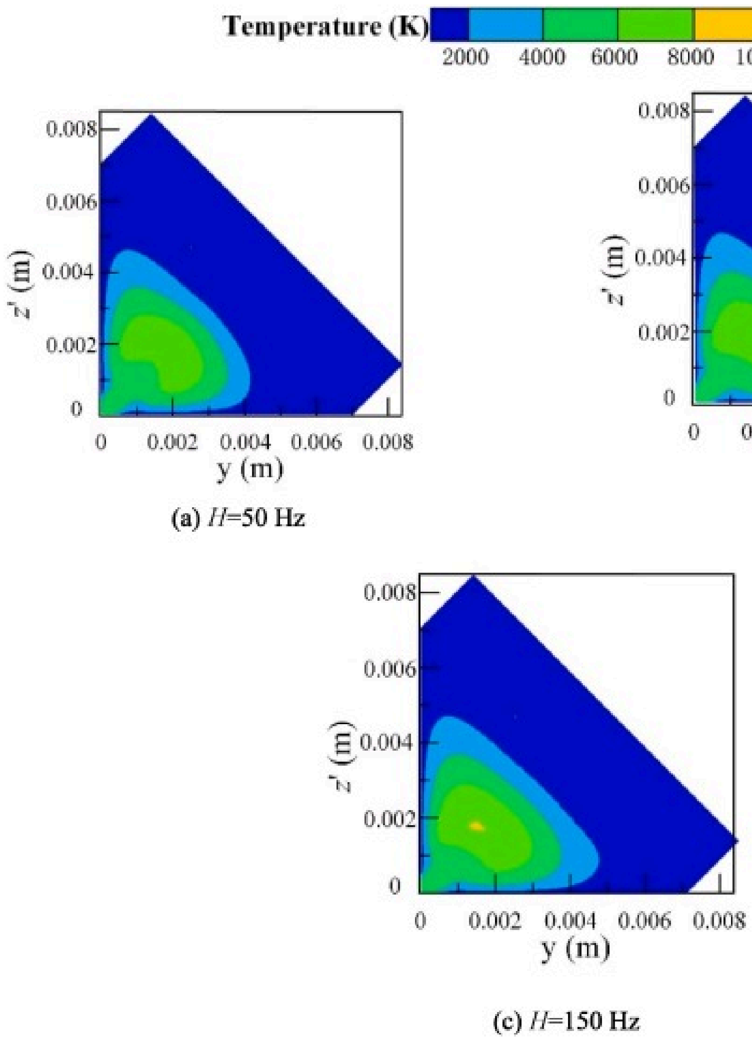

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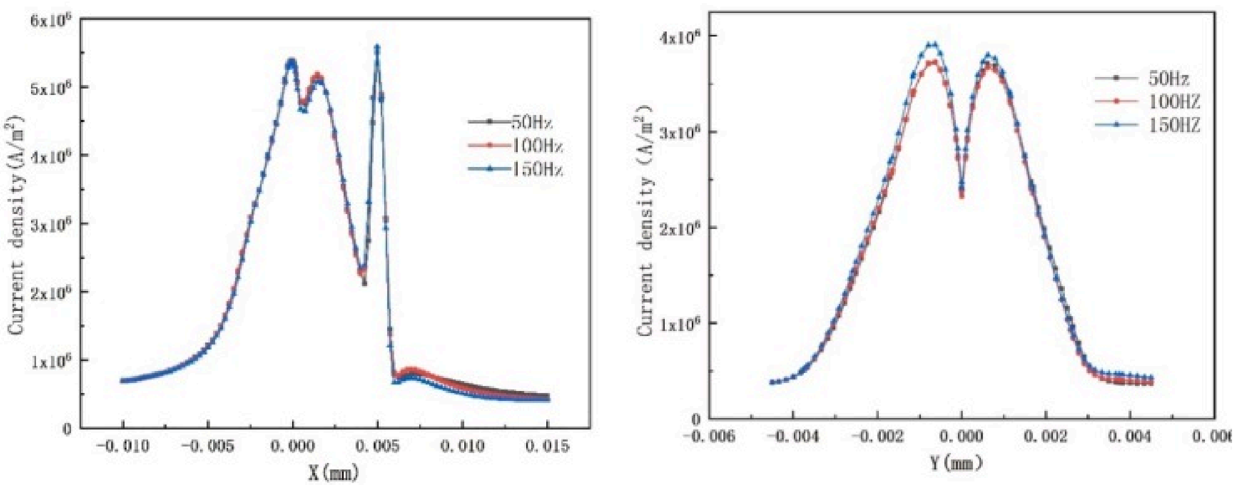
Fig. 18. Cross-section temperature field at position B.

4.3.2. Current density

As frequency increases, the workpiece surface peak current density gradually increases as shown in Fig. 19. This is because, as frequency increases, the mass of metal vapor diffused in the arc space increases which replaces the ionization of some of the argon atoms. However, if the rotation frequency is too large, the laser beam moving speed is too fast, while the amount of molten metal per unit time, metal evaporation, and keyhole current density are reduced, the ability of the laser beam to compress the arc root is weakened and the arc energy is dispersed which eventually lead to a drop in temperature. Therefore, the change in arc temperature depends on the competitive results of these two effects, which also explains the small rise in peak temperature of the temperature field.



(a) Along the x-axis (b) Along the y-axis

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Fig. 19. Workpiece surface current density distribution under different frequencies.

4.3.3. Concentration field distribution of metal vapor

Fig. 20 and Fig. 21 show the concentration field distribution of metal vapor at different rotational frequencies. When the laser beam rotates to the B position, the rotation frequency increases, the time that the laser beam acts on the molten pool per unit time decreases, the heat input received by the workpiece decreases, while the kinetic energy and metal vapor mass sprayed from the keyhole decreases, thereby lowering the height at which the arc plasma meets the laser-induced metal. Meanwhile, when the laser beam rotates to the D position, the rotation frequency increases, the distance decreases, and the arc plasma hinders the laser-induced metal vapor, thereby limiting the metal vapor height. With the increase of rotation frequency, the confinement effect of arc plasma on laser-induced metal vapor becomes more obvious.

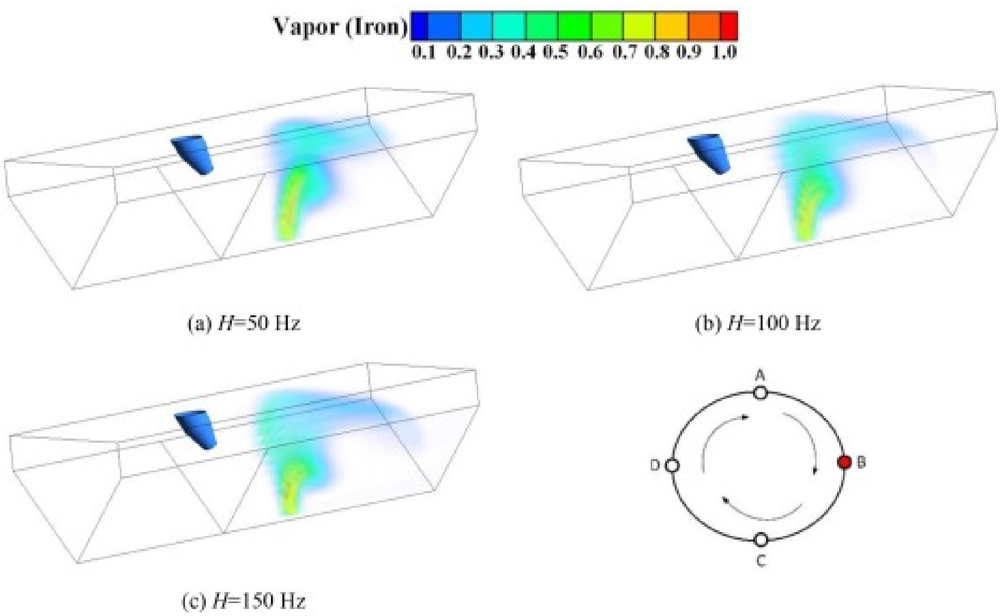

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Fig. 20. Concentration field distribution of metal vapor at position B.

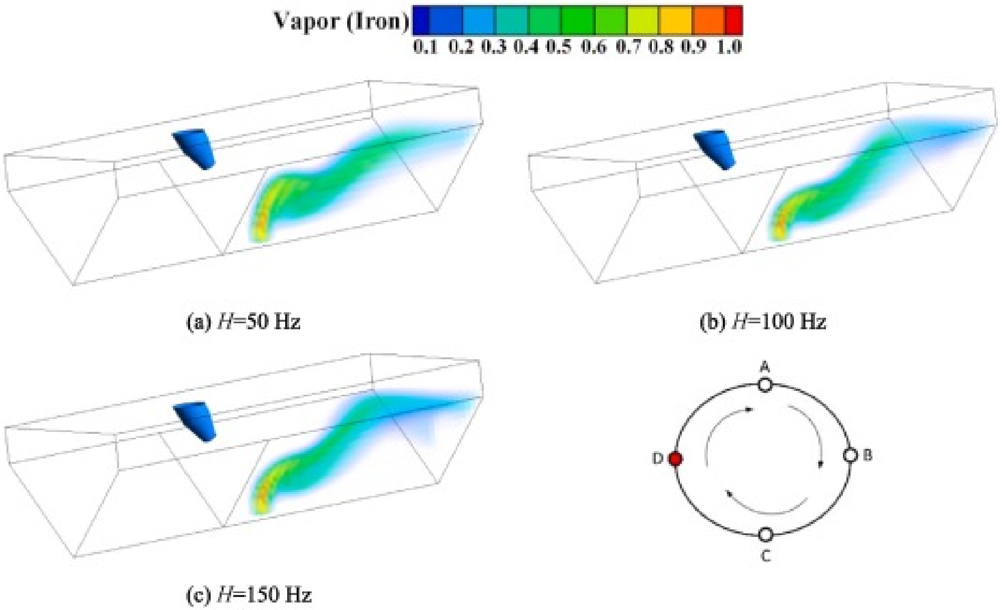

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Fig. 21. Concentration field distribution of metal vapor at position D.

4.3.4. Velocity field

The hybrid plasma velocity distribution at different rotational frequencies is shown in Fig. 22. When the laser beam is at the D position and the rotation frequency is 50Hz, 100Hz, and 150Hz, the speed peaks all appear at the end of the welding wire which are 214.31 m/s, 220.64m/s, and 229.54m/s, respectively. A clockwise eddy current appears on the right side of the laser beam action area. The higher the rotation frequency, the closer the eddy current center is to the workpiece surface. This is because when the laser beam rotates to the D position, the metal vapor and arc plasma meet at a relatively close position wherein their interaction is apparent; as the frequency increases, the metal vapor kinetic energy ejected from the keyhole decreases and has a negative effect on the arc plasma. The hindering effect is enhanced, and the right deflection height is reduced after the arc plasma and laser-induced metal vapor are mixed.

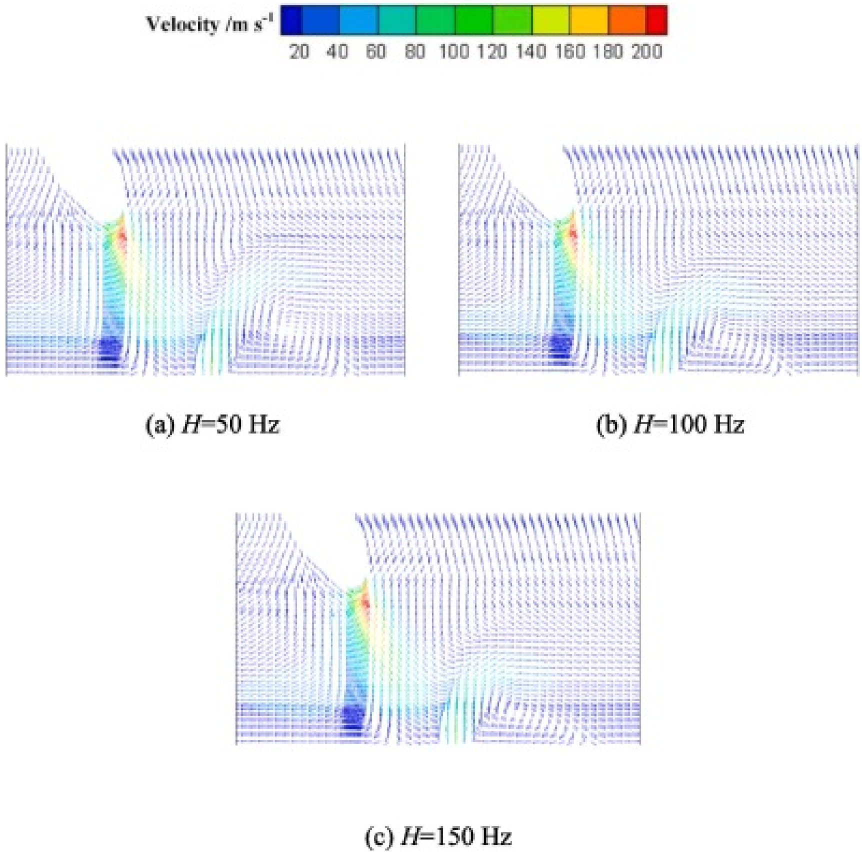

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Fig. 22. Velocity field of the hybrid plasma at position D.

4.4. Effect of rotation radius

4.4.1. Temperature distribution

The hybrid plasma temperature distribution on longitudinal sections at different positions and under different rotation radii is shown in Fig. 23. Fig. 23(a) shows the hybrid plasma temperature distribution in the longitudinal S1 section under different rotation radii when the laser beam is rotated to the B position. When the rotation radius is 0.5 mm, 1 mm, and 2 mm, the meeting heights of metal vapor and arc plasma decrease at 2.85 mm, 2.43 mm, and 2.1 mm, respectively. This is because with the increase in rotation radius, the laser rotation speed increases accordingly. Under the same rotation frequency, the action time of the laser in the same position as the workpiece decreases which affects the laser-induced keyhole effect and reduces the injected metal vapor kinetic energy while its rising height decreases. The shape of the arc on the left side of position B has little difference, while with the increase in rotation radius, on the right side of position B, hybrid plasma plume volume becomes larger and the laser compression effect on the arc gradually weakens. Fig. 23(b) shows that when the laser beam rotates to the C position, the maximum arc temperature corresponding to the rotation radius of 0.5 mm, 1 mm, and 2 mm are all located at the end of the weld which are 22183 K, 22178 K, and 22173 K, respectively, with little difference in peak values. This is because the larger the rotation radius, the shorter the action time between the laser and workpiece, the lower the laser-induced metal vapor kinetic energy, and the C position is far from the arc center so the effect on the welding wire axial direction cross-section (S2 section) is small. Fig. 23(c) shows the hybrid plasma temperature distribution in the longitudinal S1section under different rotation radii while the laser beam is rotated to the D position. With the increasing rotation radius, the keyhole position approaches closer to the end of the welding wire which has an increasing influence on the central area shape of the arc. When the rotation radius is 2 mm, the hybrid plasma is unstable and the coupling effect of the arc and laser heat source is poor. This is because there is an optimum range of spacing between the hybrid welding wire tip and the laser beam. When the laser beam action range is not within the optimal laser-arc distance range, the hybrid plasma temperature will decrease.

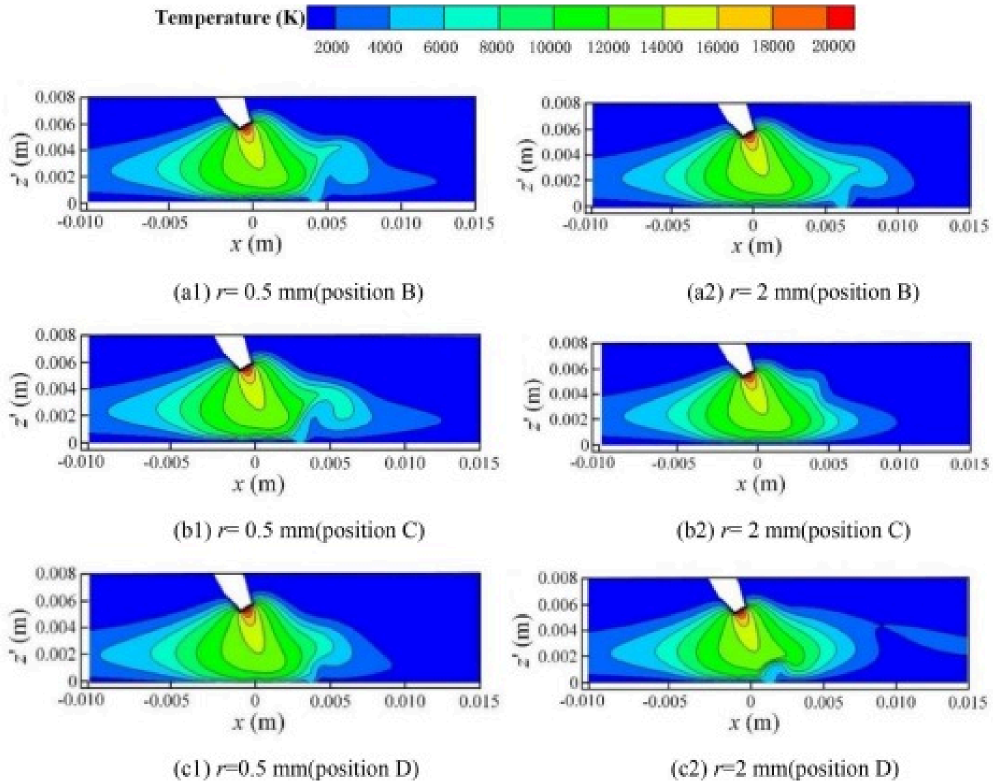

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Fig. 23. Longitudinal sections temperature field at different positions.

Fig. 24 shows the temperature distribution of the hybrid plasma in S2 cross-section at the C and D positions under different rotation radii. When the laser beam rotates to the C position, the distance between the laser beam and arc center is far which does not affect the S2 cross-section temperature distribution. When the rotation radius increases to 2mm, the laser beam rotates to the D position and the laser-induced metal vapor directly affects the central area temperature of the arc.

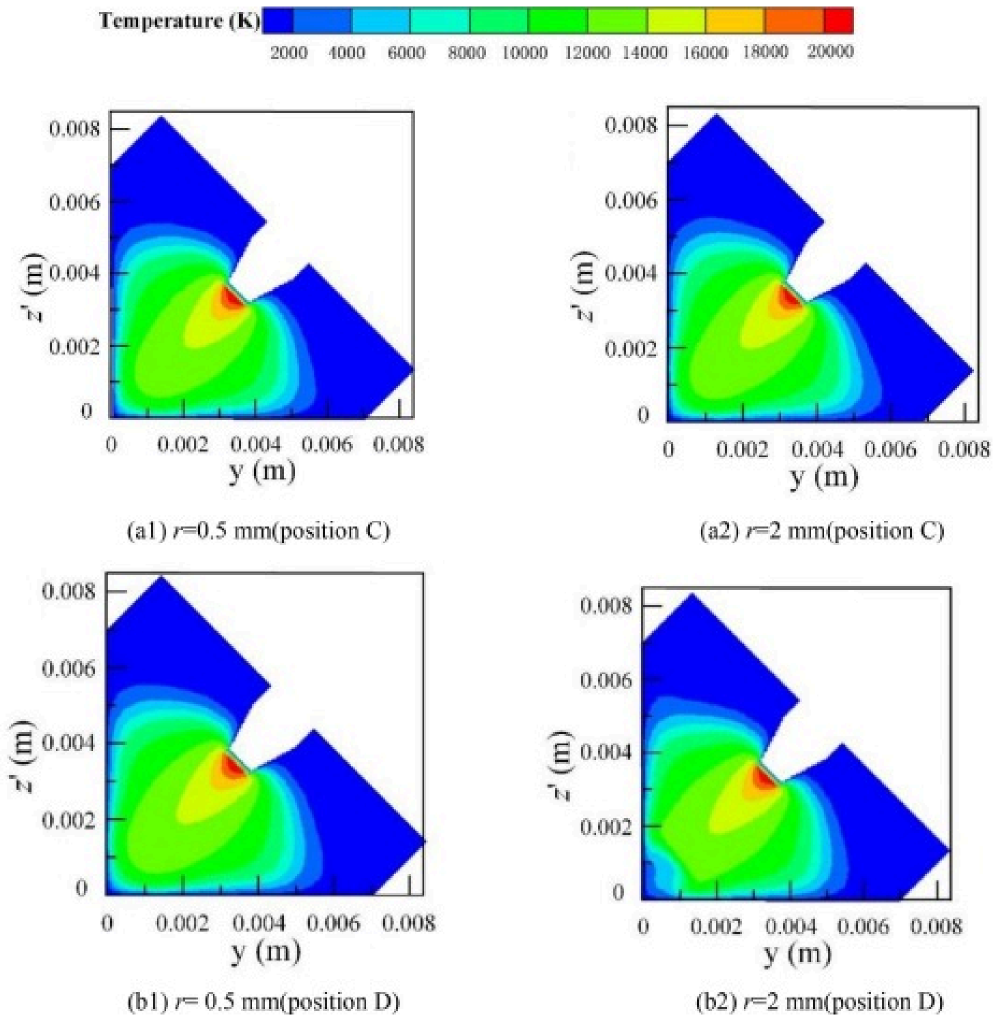

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Fig. 24. Cross-sections temperature field at different positions.

In order to quantitatively analyze the rotation radius influence on the hybrid plasma temperature, the hybrid plasma temperature near the base metal workpiece surface under different rotation radii was compared as shown in Fig. 25. When the laser beam rotates to the B position, the keyhole area temperature near the workpiece surface rises within the metal vapor action range while the “concave-shaped” transition area is less concave. Also, when the laser beam rotates to the D position, the metal vapor acts. Within the range, the “concave-shaped” transition zone is concave to a greater extent, and its position is also close to the end of the welding wire. This is mainly caused by the different degrees of cooling of the arc caused by the laser-induced metal vapor rotating to different positions.

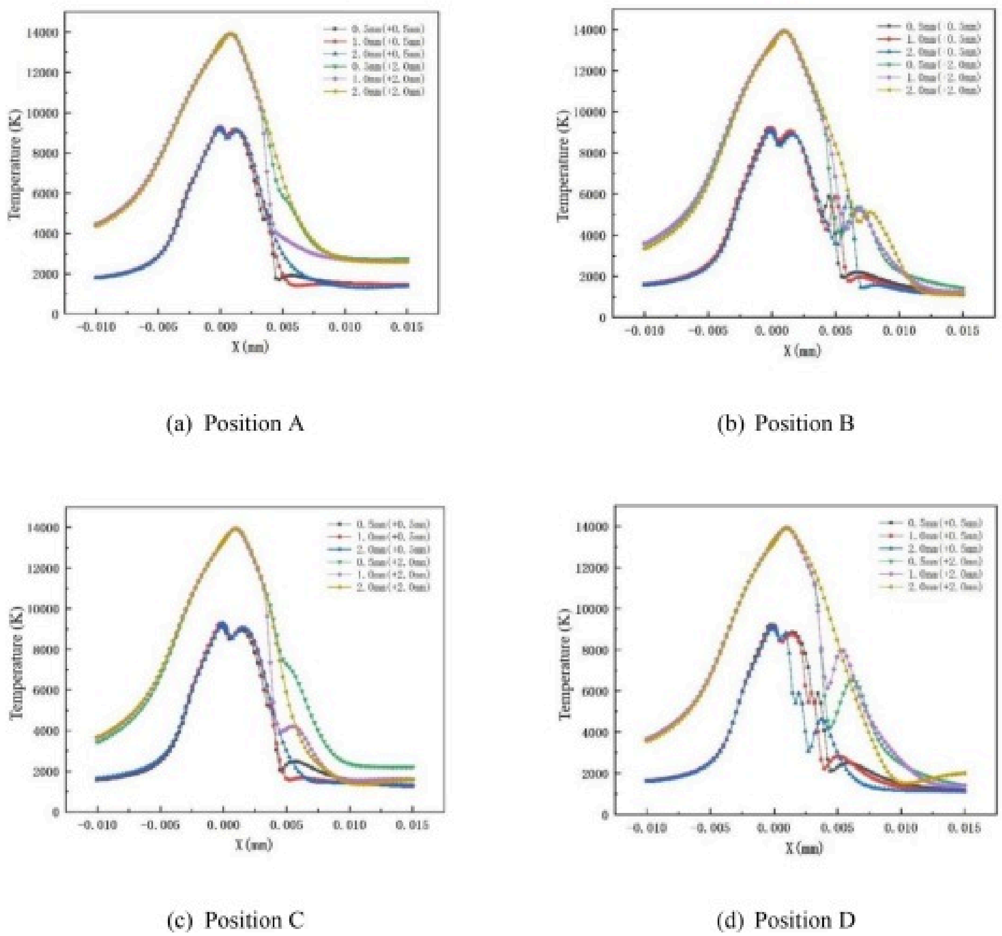

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Fig. 25. Temperature distributions of hybrid plasma at the surface of workpiece.

4.4.2. Current density

Fig. 26 shows the workpiece near-surface y-axis direction current density curves under different rotation radii when the laser beam is at different positions. The red dashed line represents where the current density was taken. When the laser beam is at the A or C position in the T-joint, the arc is more likely to start on the bottom plate and vertical plate, the plasma deviates from the welding wire axis, and the current density decreases at the angle between the bottom plate and vertical plate (Y=0). Then, showing a “concave shape,” the current density at the keyhole position changes abruptly. When the rotation radius is 0.5 mm and 1 mm, the current density distribution is in bimodal form. Meanwhile, when the rotation radius is 2 mm, it is in trimodal form. This is because when the laser beam is at the A or C position, the rotation radius is too large, more metal charged particles are ionized at the keyhole, and the proportion of argon ions participating in the ionization gradually becomes smaller. With the increase in rotation radius, the current density on the side without laser action does not change much, while on the side with laser action, the arc space current at the angle between the bottom plate and vertical plate decreases. When the laser beam is at the B position, far from the arc center, the overall current density is smallest. Meanwhile, when the laser beam is at the D position, it is closest to the arc center, and the overall current density is largest.

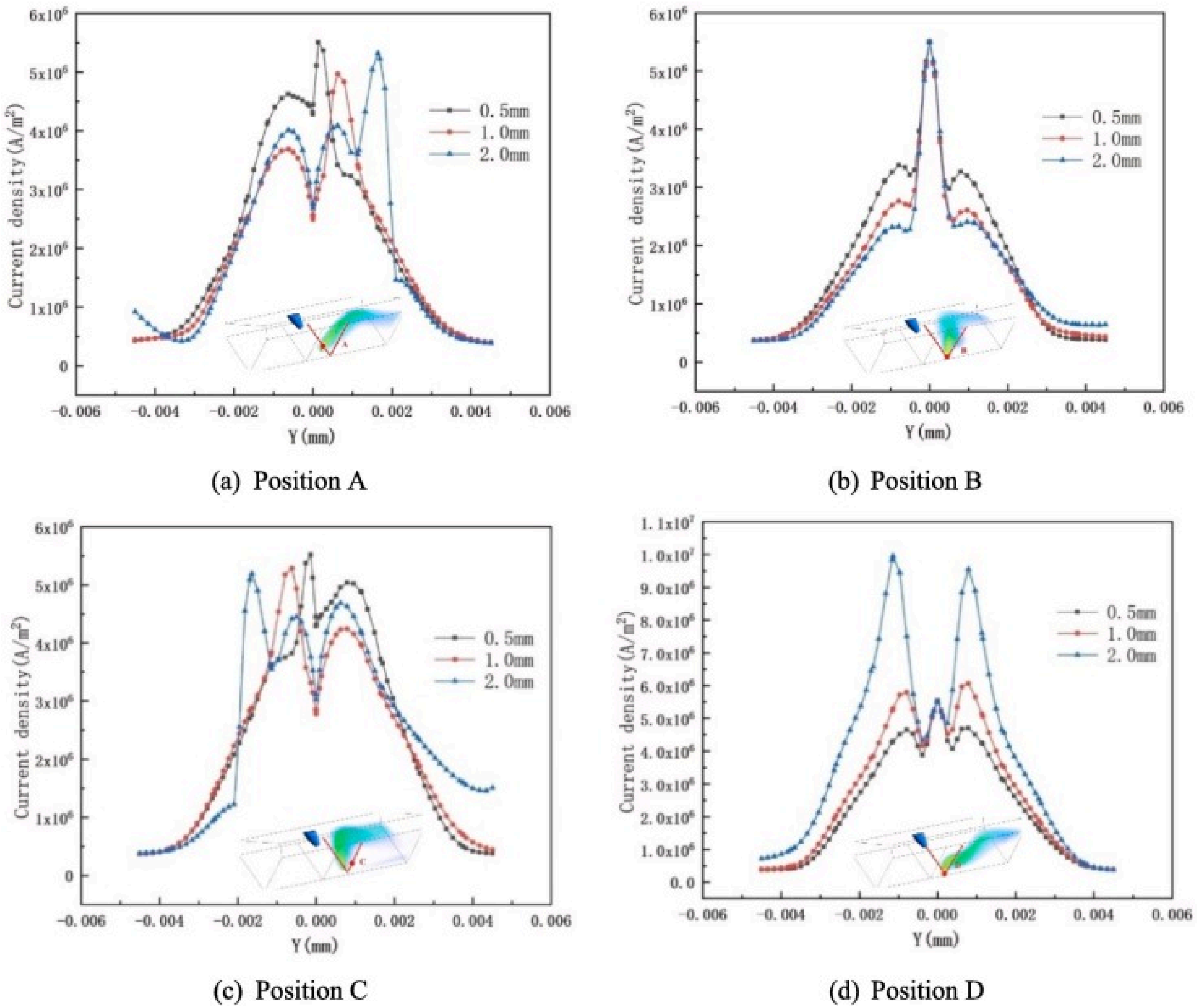

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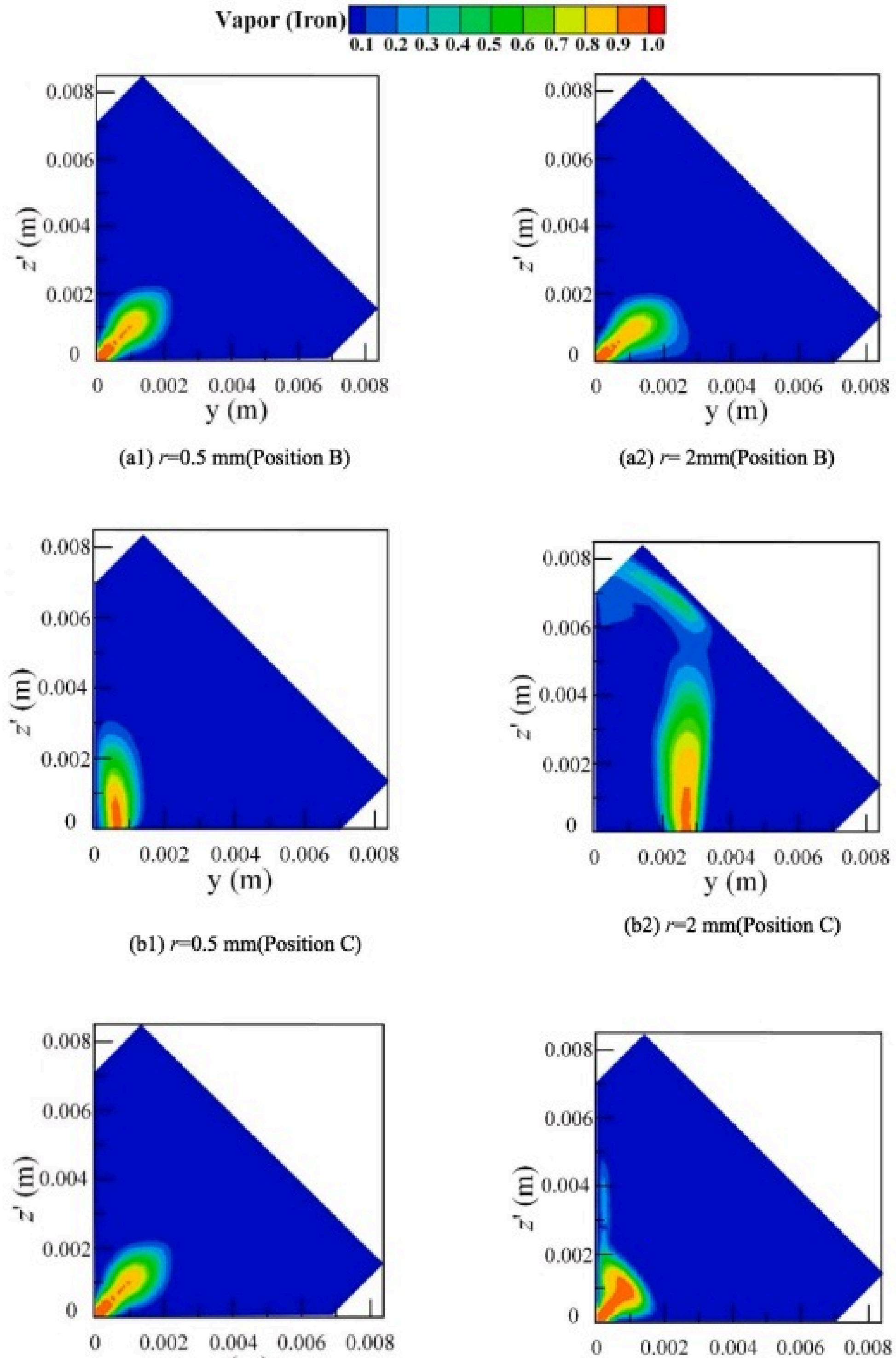
Fig. 26. Workpiece surface current density distribution along the y-axis at different positions.

4.4.3. Concentration field distribution of metal vapor

Fig. 27(a) and (c) show the concentration field distribution of metal vapor under different rotation radii when the laser beam is at positions B and D. It can be seen that as the rotation radius increases gradually, the height achieved decreases gradually when the metal vapor concentration is 0.6. This is because with the increase in rotation radius, the metal vapor eruption speed gradually decreases and the closer the laser-induced keyhole is to the welding wire end, causing greater impact from the arc plasma so that the height the metal vapor can reach is limited. When the rotation radius is 0.5 mm and the laser beam is at the four positions of A, B, C, and D, the overall difference of the arc shape is not large; however, when the rotation radius is increased to 2 mm, the arc tail oscillation is more obvious. When the laser beam rotates to the A position, the rotation radius increases and the keyhole position moves further away from the arc center until the interaction effect between the laser and arc disappears completely. Fig. 27(b) shows shows the metal vapor distribution cloud diagram under different rotation radii when the laser beam is at position C. When the rotation radius is 0.5 mm, the laser position is closer to the center section of the weld, and the metal vapor concentration is 0.6 while the height reached is 1.9 mm. As the rotation radius increases to 2 mm, the metal vapor concentration remains at 0.6 while the height reached is 3.8 mm.



(a1) r=0.5 mm(Position B)

(a2) r= 2mm(Position B)

(b1) r=0.5 mm(Position C)

(b2) r=2 mm(Position C)

y (m) y (m)

(c1) r=0.5 mm(Position D) (c2) r=2 mm(Position D)

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Fig. 27. Metal vapor distribution under different rotation radii at different positions.

In addition, the laser-induced metal vapor cooling effect is also different depending on the distance from the workpiece surface which corresponds to the arc temperature distribution cloud diagram under different rotation radii indicated above. In the “concave-shaped” transition zone, with the increase in rotation radius, the workpiece surface temperature also gradually increases. This is because with the continuous increase in rotation radius, the laser beam moving speed increases, the metal vapor ejection kinetic energy decreases, and the laser-induced metal vapor cooling effect on the arc is weakened, so the workpiece surface temperature in this area increases. In addition, the further away from the workpiece surface, the less the arc temperature rises.

4.4.4. Velocity field

The hybrid plasma velocity field distribution at the D position under different rotation radii is shown in Fig. 28. It can be seen that as the rotation radius increases, the closer the laser-induced keyhole is to the end of the welding wire and the more severe the laser-induced metal vapor disturbance is to the arc flow. When the rotation radius is 0.5 mm, the arc plasma moves along the welding wire axis under the electromagnetic force, and under the action of the vertically upward metal vapor, it mixes into a hybrid plasma and flows out to the upper right. The larger the rotation radius, the closer the eddy center of the hybrid plasma flow is to the workpiece surface. In addition, when the rotation radius is 2 mm and the laser beam is at the D position, the laser-induced vapor plume directly hinders the arc plasma from moving along the axis which will cause the arc plasma plume to be unstable. The laser-induced keyhole position gradually moves closer to the welding wire end and the arc plasma flow velocity increases continuously. As the rotation radius increases, the laser-induced metal vapor eruption velocity gradually decreases. Therefore, the closer to the end of the welding wire, the larger the radius rotation, and the lower the hybrid plasma rise.

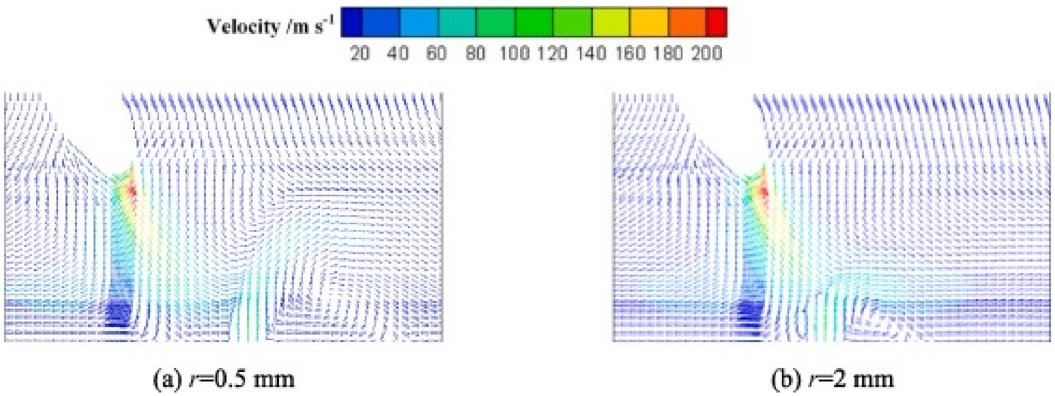

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Fig. 28. Velocity field in hybrid welding under different rotation radii at position D.

4.5. Verification of model

The spectral information acquisition system determines the plasma electron temperature by detecting plasma wavelength and intensity, which is a more accurate technique for studying plasma physical parameters. This paper uses the Boltzmann plot method[39] to determine the electron temperature. Fig. 29 shows the simulation and detection results of the hybrid plasma temperature values sequentially distributed at 1 to 7 points. The process parameters used in the experiment are: p=3 kW, I=180 A, d=2 mm, Q_{gas}=20 L/min, and v=15 mm/s. Based on the figure, the temperature rise and fall trend of the acquisition points from 1 to 7 is the same, which shows the reliability of the laser-GMAW paraxial hybrid welding arc numerical model.

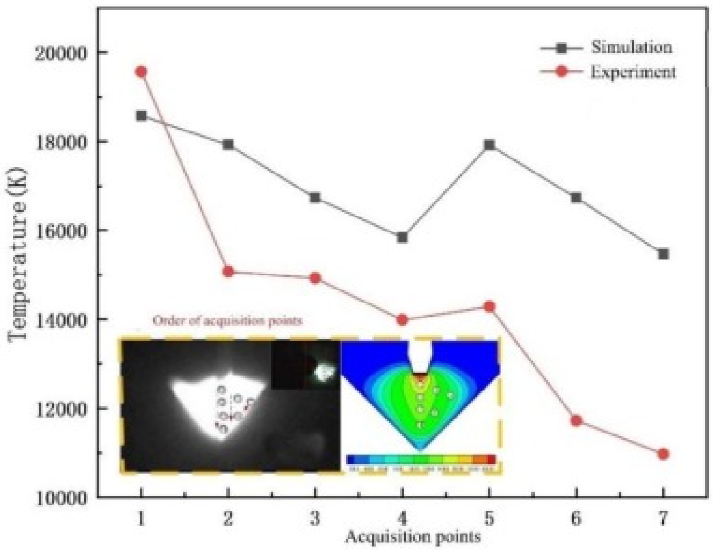

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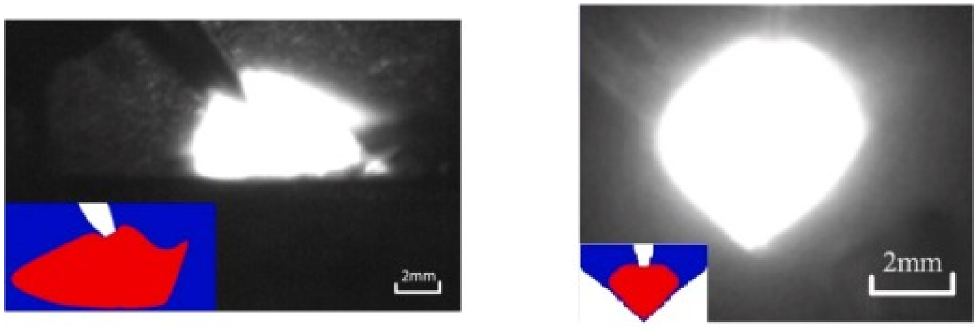
Fig. 29. Simulation and experiment 1～7 collection point plasma temperature curve.

Besides, it can be found that there still exits some divergency between the calculated and measured data. As a preliminary study, this manuscript is mainly focused on the influence laser induced metal vapor on arc dynamic behavior. Due to the complexity of physical process in rotating laser hybrid welding, the model has to be simplified to achieve the convergence of numerical calculation for coupled physical field. Similar to previous studies[23], [18], [24], [40], [41], to speed up the simulation process, the ionization process of shielding gas and metal vapor is ignored, thus causing an increase in arc temperature to some extent. Besides, the calculation error is also attributed to both the simplification of boundary conditions and lack of material properties at high temperature to some degree.

The plasma shape experimental and simulation results are shown in Fig. 30 wherein the outermost isotherm of the simulation results is taken as 6000K. Both are in general agreement, further validating the reasonability of the developed.



(a) Laser-GMAW hybrid welding longitudinal section

(b) Laser-GMAW hybrid welding cross section

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Fig. 30. Comparison of experimental and simulation arc shape.

5. Conclusions

(1). The addition of the laser provides a stable conduction channel for the arc plasma which plays the role of attracting and stabilizing the arc. Affected by the joint geometric characteristics, the arc is more likely to start on the surface of the bottom plate and vertical plate, the arc deviates from the welding wire axis, and the current density at the angle between the bottom plate and vertical plate decreases significantly. Due to the laser-induced metal vapor disturbance, the plasma temperature at the hybrid welding arc tail decreases, the flow state above the small hole area changes, and the current density becomes more concentrated.

(2). The evolution process of the typical physical characteristics of the arc is studied. The arc plasma perturbation position by the high-speed laser-induced plume ejected from the keyhole is constantly changing, and the main temperature change occurs at the arc tail. Meanwhile, the arc plasma and laser-induced metal vapor at point D have the lowest meeting height, whereas the meeting height at point B is highest. The arc effective distribution radius changes with the rotation path.

(3). The laser beam rotation frequency effect on arc characteristics is studied. The higher the rotational frequency, the stronger the arc plasma confinement effect on laser-induced metal vapor. As rotation frequency increases, the laser beam cooling effect on the arc per unit time decreases.

(4). The laser beam rotation radius influence on arc physical properties is studied. When the rotation radius increases, the laser rotation speed correspondingly increases, and the laser action time on the same position as the workpiece decreases which affects the laser-induced keyhole effect, resulting in a decrease in the sprayed metal vapor kinetic energy and height. When the rotation radius is 2 mm, the laser beam action range is not within the optimal laser-arc distance range, and the hybrid heat source coupling effect is poor.

Funding

This research is financially supported by the National Natural Science Foundation of China (Grant No. 51975263).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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